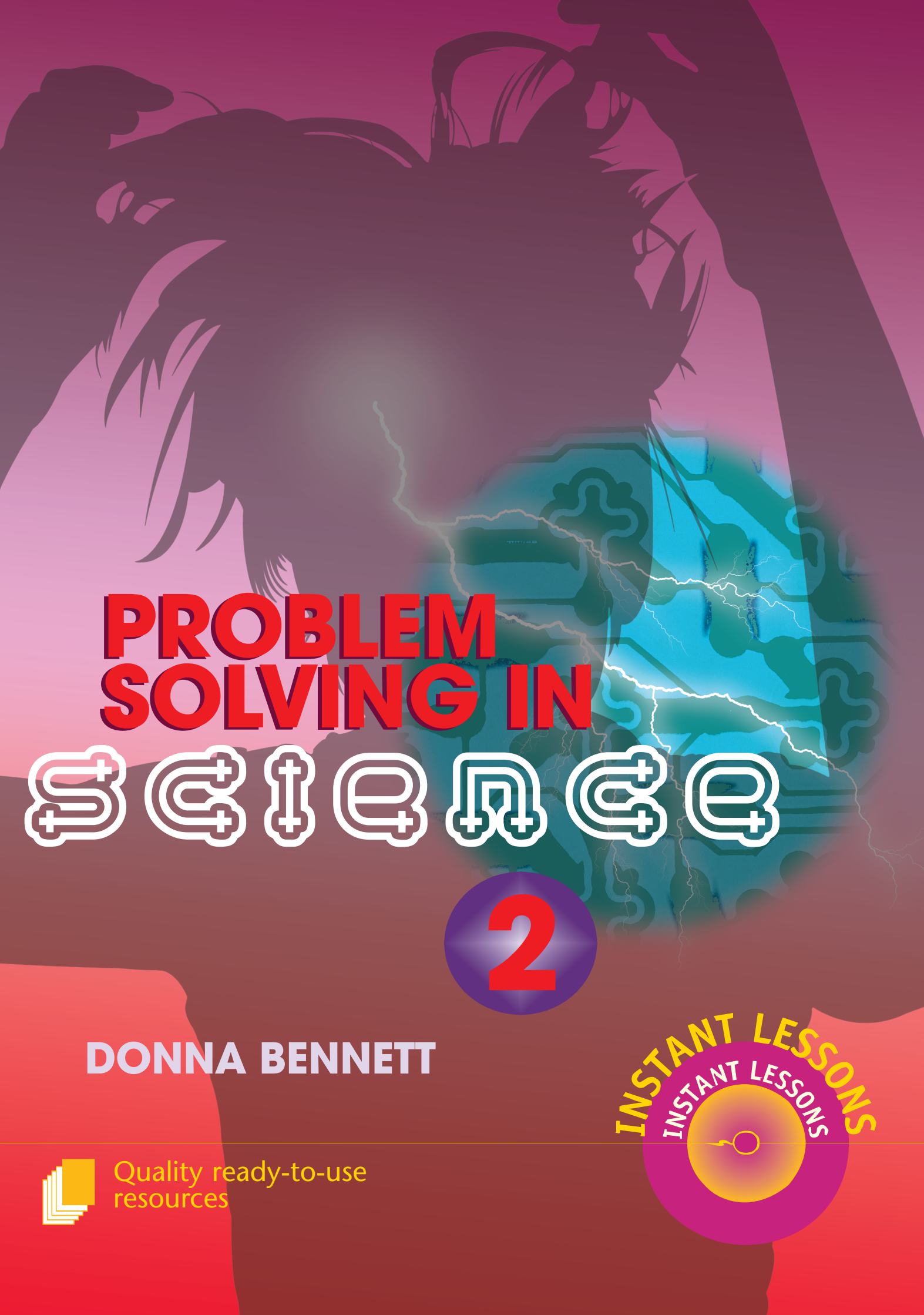




PROBLEM SOLVING IN

SCIENCE

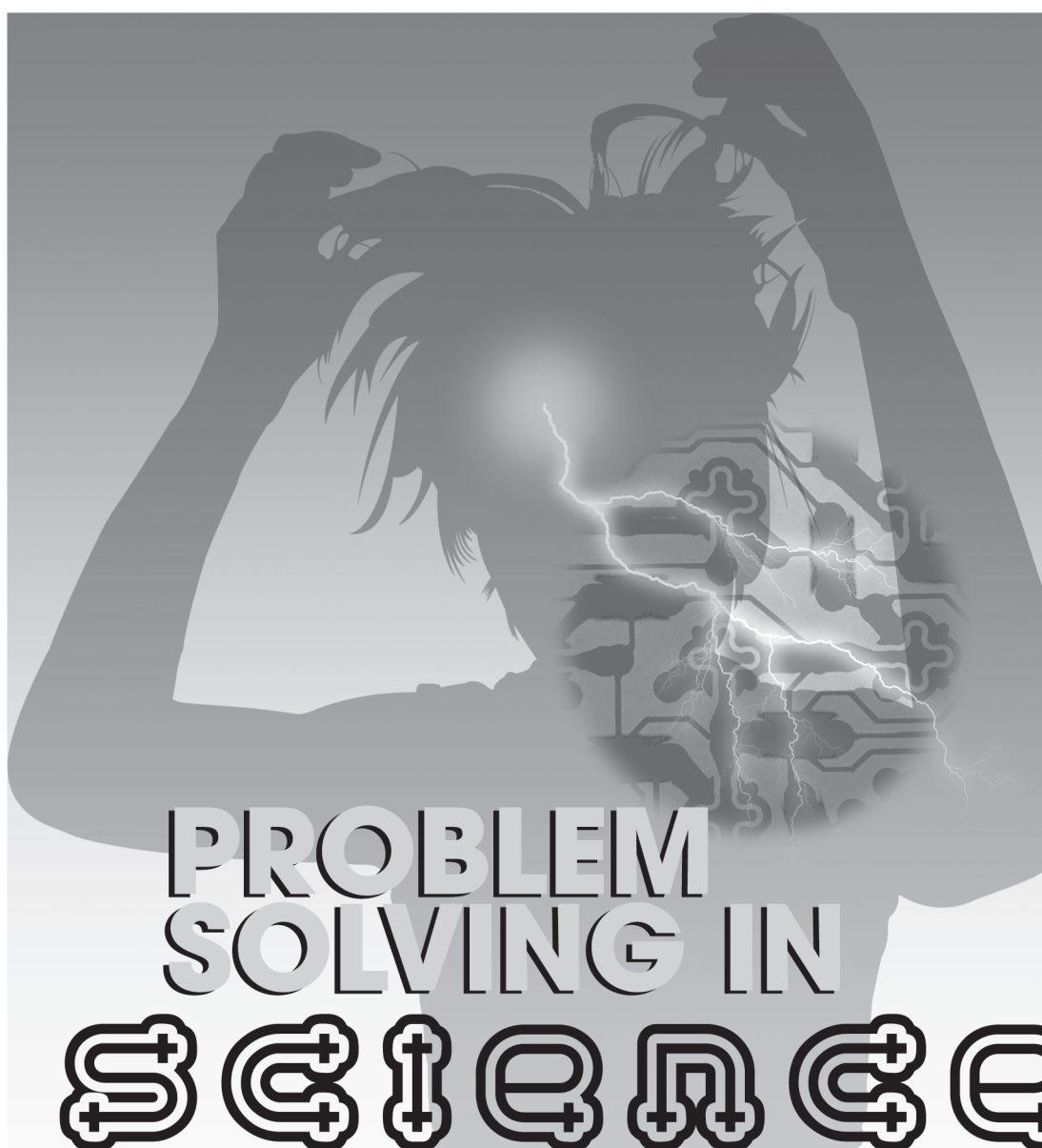
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DONNA BENNETT



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PROBLEM SOLVING IN SCIENCE

2

DONNA BENNETT





Donna Bennett is a highly qualified and experienced teacher. She is the winner of two national awards, the CRA Mining Company Fellowship and the BHP Science Teacher Award, for innovative and creative delivery of science. She is the Director of Science Education Australia and her science workshop programs are always popular with teachers and students alike. Donna has written several successful science education books, established the Queensland Science Marathon and has taken her 'Science for Bush Kids' program to regional areas. She has travelled internationally where her special form of science education has been in demand.

Donna believes that science is the greatest adventure that has ever been devised. She sees teachers as consultants and cheerleaders who provide probing questions, enthusiasm and confidence and who tease out an understanding of the basic notions of science. Teachers are a source of knowledge and information, which is shared in order to guide student exploration, rather than define it. As an author she has a clear and effective style and believes that interactive science investigations contribute most to learning when they are part of a well-thought out plan that triggers the natural curiosity of students. She provides a freedom that allows students the pleasure of building their own knowledge and understanding. Once they get over the shock of having this type of responsibility, and experience the thrill of succeeding, nothing can stop them.

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INTRODUCTION

The formulation of a problem is often much more essential than its solution, which may be merely a matter of mathematical or experimental skill. To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advances in science. (Albert Einstein)

This book helps students to become critical thinkers and independent, creative problem solvers by stimulating student thinking beyond factual recall and procedural levels. This book engages students in higher order thinking, which involves the application, synthesis and evaluation of knowledge and skills within a scientific context. For example, students will not only be able to recognise Newton’s Second and Third Laws but will understand them and be able to apply them to solve a previously unseen problem.

This book is not a laboratory manual or a series of lesson plans. This book **is not** a bunch of activities and cookbook directions to prove what students already know. However, there is a five-step order of presentation within each unit, and the teacher may interpret and adapt material to fit individual teaching agendas. Each unit begins with a **‘Curiosity trap’**, which is located in the teacher’s notes. It is recommended that this be used to grab students’ attention and arouse their curiosity before they know what the teacher has planned. This is followed by the **‘Give us a clue’** section, a hands-on activity to flesh out the basic scientific notion presented by the ‘Curiosity trap’. Next, **‘The challenge’** provides the vehicle to apply this notion to an unknown situation; it is not to be used as an example of the scientific method. Content (**‘So what’s going on?’**) plus questions (**‘Figure it out!’**) and answers are provided for each unit, to consolidate an understanding of the basic notions presented.

Using this material for a class or year level competition – A criteria sheet, which outlines a scoring system, is provided on page 5. There are nine criteria that are generic to all challenges, but the tenth criteria is specific to each challenge. This tenth criteria for each unit is provided on page 6.

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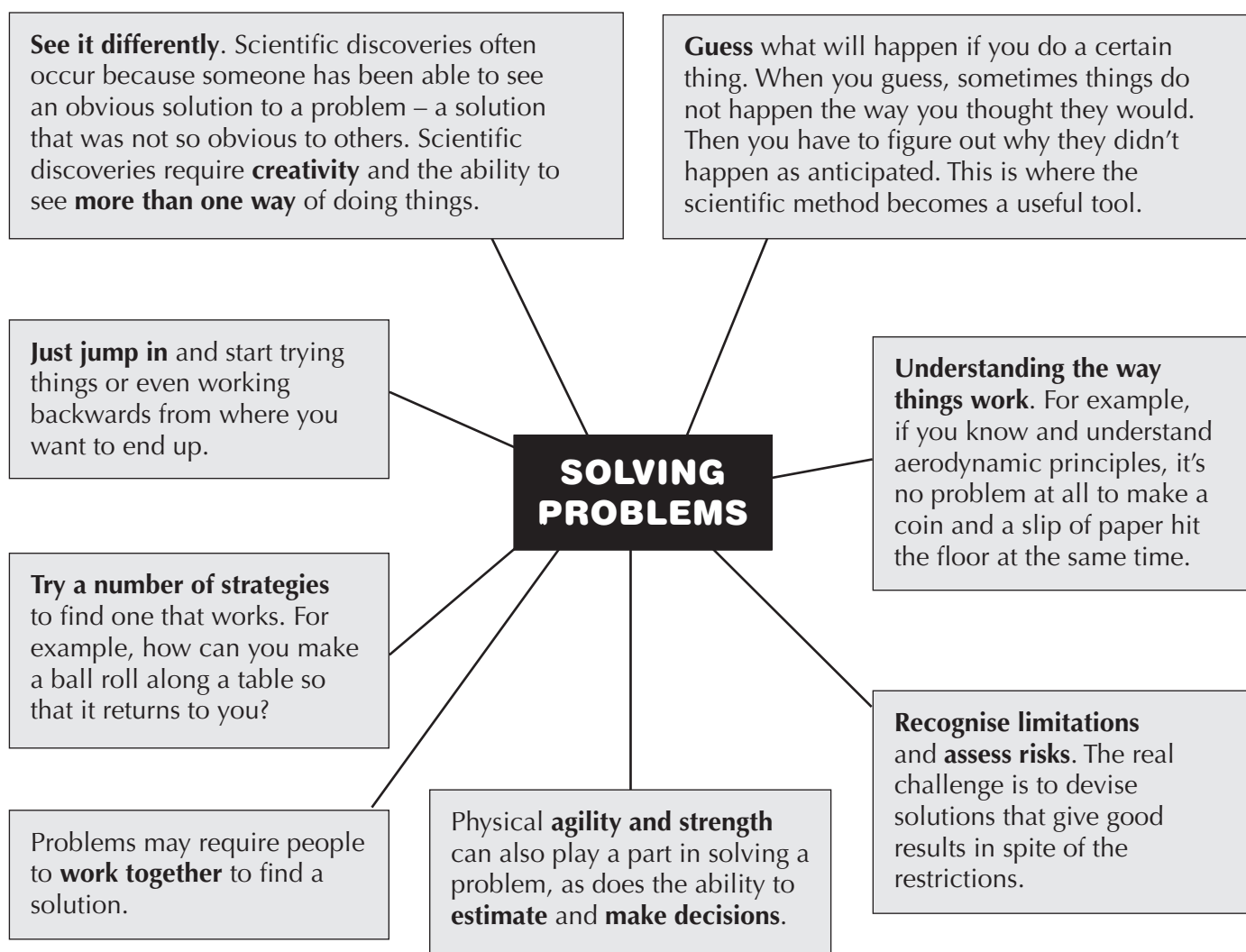
SCIENCE INVOLVES PROBLEM SOLVING

The scientific method and problem solving can stand alone, but knowledge and skills cannot.

Knowing something or knowing about something leads to the raising of a question. The Scientific Method is a great tool to use to find out the answer to the question. Once this new knowledge is accepted, it can then be used to help solve the original problem. We learn nothing unless we proceed from the known to the unknown.

Skills have limited problem solving value in the absence of the ability to use them knowledgeably, just as knowledge has limited problem solving value in the absence of the ability to apply it skilfully. Day-to-day learning should ensure that knowledge and skills are gained simultaneously.

When you want to solve a problem, you try to remember things that you know or things you can do that might help. You look for new information about the question. You may also ask yourself what you know about the situation, what more you need to know, and what assumptions you are making.



The last word

When you try to solve a problem, you try to remember things that you know or can do that might help you. You look for new information about the question. You may also ask yourself what you know about the situation, what more you need to know and what assumptions you are making. Being wise is not only about knowing and understanding lots of things, it's also about applying them appropriately.

CRITERIA SHEET

Teacher's notes

TEAM: _____ **ROOM:** _____ **FINAL SCORE:** ____ /100 **JUDGE:** _____

To score: For each criteria, circle ONE score only, plus the applicable penalties. Then transfer the circled scores to the subtotal. Subtract the total penalties from the initial score and record the actual score in the box. To calculate the final score, add up all the scores in the boxes.

Criteria 1: Team cooperation	Score	Penalty
All team members working cooperatively all the time	10	
All team members working cooperatively most of the time	7	
All team members working cooperatively sometimes	5	
One team member dominating		2
One team member was left out / did not take part		2
Team displaying uncooperative behaviour		2
Subtotal		
TOTAL (score – penalty)		

Criteria 2: Systematic planning	Score	Penalty
Demonstrating strong systematic planning	10	
Demonstrating systematic planning at all times	7	
Little systematic planning	5	
No systematic planning demonstrated		2
Subtotal		
TOTAL (score – penalty)		

Criteria 3: Knowledge	Score	Penalty
Strong evidence of useful knowledge being used	10	
Some evidence of knowledge being used	7	
Little prior knowledge being used	5	
No evidence of prior knowledge being used		2
Subtotal		
TOTAL (score – penalty)		

Criteria 4: Applying strategies	Score	Penalty
Consistently displaying use of strategies to trial and test ideas	10	
Sometimes displaying strategies to trial and test ideas	5	
No clear display of strategies to trial and test ideas (ad hoc approach)		2
Subtotal		
TOTAL (score – penalty)		

Criteria 5: Manipulating and using equipment	Score	Penalty
Excellent manipulation skills	10	
Sound manipulation skills	5	
Inappropriate use of equipment		2
Subtotal		
TOTAL (score – penalty)		

Criteria 6: Safe techniques	Score	Penalty
Demonstrating safe and responsible techniques at all times	10	
Demonstrating safe and responsible techniques sometimes	5	
Demonstrating unsafe or irresponsible techniques		2
Subtotal		
TOTAL (score – penalty)		

Criteria 7: Innovation	Score	Penalty
Demonstrating innovative/creative thinking in solving problems	10	
Demonstrating ordinary/simplistic ways of solving problems	5	
Subtotal		
TOTAL (score – penalty)		

Criteria 8: Attention to task	Score	Penalty
Focused and persistent; constantly on task	10	
Some interruption to concentration	5	
Distracted and inattentive (one or more team members)		2
Subtotal		
TOTAL (score – penalty)		

Criteria 9: Application of basic scientific notions	Score	Penalty
Application of the relevant basic scientific notion	10	
Some evidence of application of the relevant basic scientific notion	5	
No evidence of application of the relevant basic scientific notion		2
Subtotal		
TOTAL (score – penalty)		

CRITERIA 10 SCORING

Teacher's notes

1. What a muddle!	Score	Penalty
Correctly identified 5 products	10	
Correctly identified 4 products	7	
Correctly identified 3 products	5	
Correctly identified 2 products	3	
Correctly identified 1 product	1	
No products correctly identified		2
Subtotal		
TOTAL (score – penalty)		
2. Jacuzzi jigsaws	Score	Penalty
The hydraulic machine successfully turned on the buzzer	10	
The hydraulic machine worked but did not turn on the buzzer	5	
The hydraulic machine malfunctioned (loose connections)		2
Subtotal		
TOTAL (score – penalty)		
3. Domino Disaster	Score	Penalty
Domino 'topple' time		
More than 5 seconds	10	
Between 4 and 5 seconds	8	
Between 3 and 4 seconds	6	
Between 2 and 3 seconds	4	
Between 1 and 2 seconds	2	
Did not topple or failed in some way		2
Subtotal		
TOTAL (score – penalty)		
4. Bubble busters	Score	Penalty
Criteria A:		
Biggest bubble bigger than a fully inflated balloon	5	
Biggest bubble smaller than a fully inflated balloon	3	
Biggest bubble smaller than a tennis ball	1	
Unable to blow a bubble for testing		2
Subtotal		
Criteria B: Descent time		
Developed an appropriate measuring method and measured the size of the bubble accurately	5	
Developed an appropriate measuring method but could not measure the bubble accurately	2	
Could not develop an appropriate measuring method		2
Subtotal		
TOTAL (score – penalty)		
5. Getting a lift	Score	Penalty
Criteria A:		
Toy not broken	5	
Toy cracked or in some way damaged, but still usable	2	
Toy broken and unusable		2
Subtotal		
Criteria B: Descent time		
More than 10 seconds	5	
Between 5 and 10 seconds	3	
Less than 5 seconds	2	
Parachute failed to work at sometime during descent		2
Subtotal		
TOTAL (score – penalty)		

6. Explosion control	Score	Penalty
Criteria A: Distance travelled		
Travels more than 10 m	10	
Travels 8–10 m	8	
Travels 6–8 m	6	
Travels 4–6 m	4	
Travels 1–4 m	2	
Travels less than 1 m		2
Subtotal		
TOTAL (score – penalty)		
Criteria B: Rate of travel		
This calculation should be completed by, or at least checked by, the judge. The fastest rate scores 10 and all other teams lose 1 point for every m/s less than the fastest time.		
Rate of travel = $\frac{\text{Distance travelled (m)}}{\text{Time taken (s)}}$ = rate in m/s		
7. The truth is out there	Score	Penalty
Criteria A: Scientific method		
Able to analyse unknown powders using a recognised scientific process and reach a decision	5	
Analysed powders and reached a decision but did not use scientific process	2	
Analysed powders in a methodical way and reached a conclusion	1	
Unable to analyse powders or reach a reasonable decision		2
Subtotal		
Criteria B: Robber's identity		
Correct identification	5	
Incorrect identification		2
Subtotal		
TOTAL (score – penalty)		
8. Value of volumes	Score	Penalty
Criteria A: Density calculations		
Densities calculated correctly and column labelled correctly	5	
Densities calculated correctly but column not labelled	4	
Density calculations not attempted/incorrect		2
Column not labelled		1
Subtotal		
Criteria B: Substance grouping		
Substances correctly grouped for density	5	
Some groupings correct for density	3	
No part of the conclusion is correct		2
Subtotal		
TOTAL (score – penalty)		
9. How fast can you go?	Score	Penalty
Reaction rate		
Team with the largest balloon	10	
Team with smallest balloon		2
All other teams (balloon size between largest and smallest)	5	
Subtotal		
TOTAL (score – penalty)		
10. Zygoop	Score	Penalty
Feasible design based on relevant properties	10	
Design loosely connected to relevant properties	5	
Weak connection between design and properties	2	
Non-feasible design and/or non-relevant properties		2
Subtotal		
TOTAL (score – penalty)		

★ You're a chemist and you don't even know it!

The students will use organised tests to work out the nature of an unknown substance. Ask them to consider the best order in which to conduct the experiments, whether the order really matters, and how they will record their findings.

Red cabbage juice can be made by boiling the shredded leaves of a red cabbage in 3–4 litres of

water, then cooling the mixture and squeezing all the purple water into a container. This indicator will keep for several days in the refrigerator or indefinitely in the freezer. Alternatively, Universal Indicator can be used.

Set up each workstation as described for 'The challenge' below.

★ The challenge

The students will use various tests to identify as many powders as possible. Suggested white powders are cream of tartar, bicarbonate of soda, fine table salt, icing sugar, plain flour. Set up individual work areas with all of the equipment required for each team or group. The five 'white powders' should be put into containers labelled 1 to 5. Stress that only small amounts of test powders should be used and that the test samples must not be contaminated.

To reduce 'clean up' operations, cover each work area with newspaper. It is useful to provide an ice-cream

container for liquid waste and a bucket of warm water for rinsing the equipment.

To extend this concept further, other powders may also be used: crushed antacid tablets, artificial sweetener, chalk dust, cornflour, self-raising flour, instant mashed potatoes, powdered milk, onion powder, unflavoured gelatine, white sand, talcum powder.

LINKS TO THE EVERYDAY – cleaning products, indigestion tablets, limestone caves, salad dressing, cooking, stain removers, cleaning products

★ Curiosity trap

Make two pikelet mixtures, one with self-raising flour and one with plain flour. The basic mixture is made by combining half a cup of plain or self-raising flour, 1 egg, 1 tablespoon of sugar and half a cup of milk. Cook both mixtures at the same time in the same greased electric fry pan. Make the individual pikelets about 4 cm in diameter, compare them as they cook, and then have students do a taste test. Why do cooks

use self-raising flour? Raising agents such as baking powder contain an acid and a base, which react when wet (in this case, when milk is added) and produce bubbles of carbon dioxide that make the pikelet or cake light and fluffy. Self-raising flour is therefore preferred to plain flour for these particular properties. Purchase a small cake and cut it open to demonstrate the bubbles that were cooked into the cake.

★ Answers

1 **a** vinegar test **b** sugar test **c** iodine test **d** acid/base test **e** observation test **f** water test

2 **a** neutral (=7), acidic (less than 7), basic (more than 7) **b** pH = 3 **c** pH = 13

3 **a** Test each product with an indicator such as red cabbage juice or Universal Indicator **b** an indicator, a test container and stirrer for each product, a little water **c** Record the pH of each product in a data table, then analyse the results and sort the products into the following groups: acids, bases and neutral

4 When you lick sherbet the citric acid dissolves in your saliva and reacts with the bicarbonate of soda to produce the gas carbon dioxide. The bubbles tickle your tongue

5 The antacid tablet is a base and reacts with the acid in the stomach, neutralizing the surplus. Because bubbles of carbon dioxide are produced, people often burp after taking these tablets

6 **a** energy, atoms and the space between them **b** periodic table **c** solid, liquid and gas **d** The Earth is more or less a closed system, so we cannot get any more atoms. In the future we may be able to invent new types of atoms or bring them back from another part of the Universe

7 The ice cubes went pink because of the acid in the lemonade

8 **a** carbon dioxide **b** fine bubbles fixed in the bread

9 The flame is extinguished because the carbon dioxide produced deprives the flame of oxygen. Fire extinguishers work the same way

10 **a** filtration + evaporation **b** filtration **c** sieving + evaporation **d** flotation **e** filtration

★ Give us a clue – you're a chemist and you don't even know it!

What you need

A variety of fine brown powders, such as cocoa, instant coffee, cinnamon, chocolate cake mix, brown sugar, gravy base, nutmeg; magnifying glass; sheet of white paper; water; measuring cup; small beakers or drinking glasses; spoons (for mixing); eye dropper; ziploc bags; vinegar; tincture of iodine; baker's yeast; red cabbage juice; warm water

What to do

The powders may look the same, but a little detective work can show up big differences.

Label all the powders to be tested. For each powder, your group should complete all of the following tests. Draw up a data table to record the reactions of the powders (their properties).

Tests:

- 1 Observation:** Put a small amount of each powder on the white paper and have a good look at it using the magnifying glass. Rub each powder between your fingers to check for graininess.
- 2 Water test:** Put a drop or two of water on each of the powders. Does it soak right in or does it bead and roll off? If gas bubbles are produced, the powder may contain an acid and a base that react together to produce carbon dioxide.
Add a little of each powder to a glass of water. Does it float, sink or dissolve?
- 3 Iodine test:** Apply one or two drops of iodine. It will turn purple if starch is present. Flour consists of starch. The iodine will remain yellow if no starch is present.
- 4 Vinegar test:** Vinegar is a weak acid. Mix a couple of drops of vinegar with the test powder. If gas bubbles appear then the substance contains baking powder.
- 5 Acid/base test:** Add red cabbage juice to the test samples. If acid is present the liquid will be a pinkish colour, if basic it will be a greenish-blue colour.
- 6 Sugar test:** Put a spoonful of the test substance, a teaspoon of dry baker's yeast and 4 teaspoons of warm water into the ziploc bag. Squeeze out the air, seal the bag and hang it in lukewarm water for 15 minutes. The gas (carbon dioxide) produced by the yeast in the bag is proportional to the amount of sugar in the sample.

Now answer these questions:

- How are these solids the same and how are they different?
- Which substances contain an acid?
- Which substances contain starch?
- Which substances contain sugar?
- Can you organise these powders into groups – sugar, acids, starch, dissolving, etc.?

What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Properties are what a substance looks like and how it behaves.

Some powders are soluble in some liquids.

Some powders dissolve slowly and others are insoluble. Some powders form suspensions in liquids, i.e. insoluble solid suspended in a liquid.

★ The challenge

Mice have eaten the labels off the containers of baking ingredients in your kitchen. Work out which container holds which ingredient. **No tasting or smelling is allowed!**

Materials

5 containers of white powders labelled 1 to 5, red cabbage juice, magnifying glass, sheet of black paper, warm water, measuring cup, small beakers or drinking glasses, spoons (for mixing), eye dropper, ziploc bags, vinegar, tincture of iodine, baker's yeast

Instructions

- 1 You have 20 minutes to **work out** which substances are in which containers.
- 2 Use any of the equipment and materials provided in any way you wish to identify the powders.
- 3 **Record** your solution to this problem in the table below.

SAFETY!

☠ Do not taste or smell any of the ingredients.

Powder muddle

POWDER NUMBER	POWDER NAME
1	
2	
3	
4	
5	

Clues!

Cream of tartar is an acid.

Flour and sugar are neutral.

Flour is the only powder that will not dissolve.

Table salt and bicarbonate of soda are basic.

Bicarbonate of soda produces carbon dioxide when mixed with an acid.

★ So what's going on?

All the atoms in the universe have been here since time began.

Atoms, the **energy** they have and the **spaces** between them make up all the matter on Earth. All matter on Earth exists as either a solid, liquid or gas, depending on how much energy the atoms have. Scientists have had an enormous job just naming all these chemicals, let alone sorting them into groups. In 1869, a Russian scientist named Mendelyev published a list of all the different kinds of atoms on this planet. His table sorts the different atoms into groups, depending on how they behave. It is called the **periodic table** and can be found in most chemistry books.

Groupie science

Chemistry is the study of the physical and chemical characteristics of materials. Everything is made of chemicals, from the food we eat to the oceans we swim in. We depend on the interactions of chemicals and their physical characteristics every day, even to cook and to clean.

Grouping substances is a very important part of science. Scientists group substances by observing how they behave. These behaviours are called the properties of a substance. For example, one substance may dissolve in water while the other may not; one may be attracted to a magnet and the other may not.

Some properties of a substance may be immediately obvious, for example its colour, but others may not be so obvious, meaning that testing is required.

Scientists also use the specific properties of substances to separate mixtures. They exploit a property that one substance has and the others do not.

The pH factor

Chemicals are grouped together because they have similar properties. For example, **acids** are a group of chemicals that contain hydrogen. When a food tastes sour, like lime juice or vinegar, it usually contains an acid. **Bases** are the opposite of acids. They usually feel slippery, like soap, and have no really strong taste. If a substance is neither an acid nor a base, scientists call it **neutral**. The **pH scale** was invented to help scientists communicate exactly how acidic or basic substances are. The scale goes from 1, which is extremely acidic, through to 14, which is extremely basic; 7 is neutral. Acids can neutralise bases. Acids and bases will each turn an indicator such as red cabbage juice or Universal Indicator different colours. Details of these colours are given in the following chart:

COLOUR OF RED CABBAGE JUICE SOLUTION	COLOUR OF UNIVERSAL INDICATOR SOLUTION	APPROXIMATE pH NUMBER
red	red	2 (acidic)
purple	orange-red	4
violet	yellow	6
blue	greenish-blue	8
blue-green	violet	10
green	reddish-violet	12 (basic)

Raising agents in cooking

Substances called **raising agents** introduce a gas into a baking mixture giving it a lighter texture when cooked. Baking powder is a commercially prepared mixture of the base called sodium bicarbonate, plus an acid and a starch (to keep the mixture dry). Self-raising flour is plain flour to which a commercial raising agent has been added. When sodium bicarbonate is mixed with an acid – for example cream of tartar, lemon juice or vinegar – then bubbles of carbon dioxide are produced.

★ Figure it out!

- Which of the six tests available to you in the challenge would best identify the differences between the following pairs of substances?

- bicarbonate of soda and cream of tartar
- artificial sweetener and raw sugar
- chalk dust and cornflour
- self-raising flour and plain flour
- instant mashed potatoes and white sand

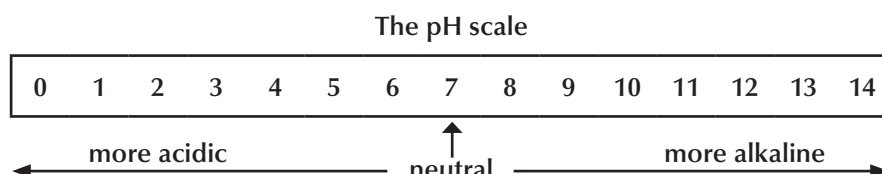
- talcum powder and powdered milk

- Study the acidity or pH scale in the diagram.

- Match the pH with the following groups – alkaline, neutral or acidic.

- Which is more acidic, pH = 3 or pH = 5?

- Which is more alkaline, pH = 8 or pH = 13?



pH	GROUP
equal to 7	
less than 7	
greater than 7	

- 3 You are given the following cleaning products: dish detergent, vinegar, household bleach, laundry detergent, shampoo, toilet cleaner, soap, window cleaner.
 - a Design an experiment allowing you to sort the products according to their acidity.
 - b List the equipment you will need.
 - c What information will you collect, and how will you record the information?
- 4 Children love to eat sherbet because it tickles the tongue. It is made of sugar, citric acid and bicarbonate of soda. Why does it behave in such a delightful way in the mouth?
- 5 Some people are greedy, eating too much food at one time; they end up suffering from indigestion. Their stomach produces a lot of acid to digest this oversupply of food, making them feel very uncomfortable. They sometimes take an indigestion or 'antacid' tablet to relieve the unpleasant feeling. How does this remedy work?
- 6 Consider the universe and what it is made of, then answer the following questions.
 - a What three things is the Earth made up of?
 - b What is the name of the chart that lists all the different kinds of atoms on Earth and groups them according to the way they behave?
 - c What are the three states of matter for materials on Earth?
 - d Can we get any more matter or any different atoms if we run out of the ones we have on Earth?
- 7 The members of a science club decided to have a party. They prepared a 'magic' drink for their guests – lemonade with purple ice cubes. The lemonade is a weak acid and the ice cubes were made by freezing red cabbage juice. What do you think happened to the colour of the ice cubes in their guests' drinks? Why?
- 8 Bakers make wonderful light and tasty breads by combining flour and sugar with a single-celled plant called yeast. Yeast grows rapidly when given warmth, moisture and food (sugar).
 - a What substance is produced in the bread dough as the yeast grows and the dough rises, making the bread fluffy?
 - b What evidence of this process can be observed in freshly baked bread?
- 9 Some students were investigating the connection between cooking cakes and how a fire extinguisher works. They knew that a fire will go out if oxygen is not present. They tested this idea by mixing some vinegar with bicarbonate of soda. They lit a match but when they held it over the bubbles the flame quickly went out. Explain this result.

10 Extend your understanding

The following physical techniques can be used to separate mixtures of solids and mixtures of solids and liquids:

- **Sieving** relies on different-sized holes.
- **Filtering** is similar to sieving and separates suspended substances from a liquid.
- **Flotation** depends on the density of liquids and solids.
- **Evaporation** of a liquid occurs when it absorbs so much energy that it turns into a gas.

This process can be used to separate substances that dissolve in water.

Which process, or combination of processes, would you use to separate the following mixtures?

- a salt and pepper
- b mud and water
- c a broken bottle of bath salts
- d a mixture of plastics
- e a mixture of water and spaghetti

★ Mouse manoeuvres

Students may build the mouse as described, or small toy mice that slide easily may be provided instead.

Syringes should always be new or otherwise well greased before use.

★ The challenge

If required, basic electrical information can be sourced from 'Burglar Alert' in Problem Solving in Science Book 1. Electronic Alarm Buzzers (70dB; Cat No L7009) and basic electrical materials can be purchased from Dick Smith Electronics. Hydraulics tend to interest male students more but this challenge

balances the interests of both male and female students.

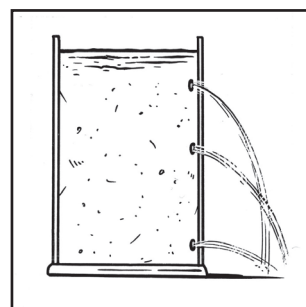
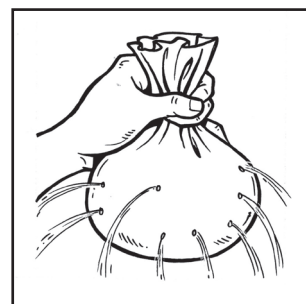
LINKS TO THE EVERYDAY – tractors, the moving arms of a backhoe and other earthmoving equipment, hydraulic car jack, hydraulic disc brakes, water pistols, hydro-electric power plants.

★ Curiosity trap

Using all three of the following ideas in the given sequence is sure to grab attention and won't take long. Take the students outside for the demonstrations and recruit one or two of them to assist.

1 Fill a medium-sized freezer bag with water, expel any air, then twist the top with your hand so that the bag is tight and the water under pressure. Maintain this twisting force at the top of the bag as you stand back and quickly poke 3–4 holes in it with a pencil. If you keep up the force on the bag, the spurts of water will all be roughly the same size for a short time. Pressure in a liquid is transmitted throughout the liquid and in all directions equally.

- 2 Take a tall tin can and punch three holes in the side, one above the other, at different depths. Cover the holes firmly with Blu Tack and fill the can to the top with water. Remove the Blu Tack. From which hole does the water squirt out the fastest and furthest? Where is the pressure greatest? Pressure increases with the depth of liquid.
- 3 Apply all this knowledge using a few cheap water pistols. They're an excellent example of a hydraulic machine and you can have a lot of fun with them.



★ Answers

- 1 a Pascal b Pressure applied to a liquid acts equally in all directions, and changes in this pressure are transmitted instantly c Depth and density of the liquid
- 2 surface D
- 3 a inclined plane, lever, wheel/pulley b They use liquids under pressure to make things happen c Liquids cannot be squashed, and pressure applied to a trapped liquid is felt throughout the liquid
- 4 a diagram B b The top one will start to move closer to the can and then stop. The middle one will move closer to the can and stop. The lowest one will be the last to do the same thing c The deeper the liquid, the greater the pressure d Yes, because the lowest hole spurts the furthest e He would get squashed by the pressure of the water molecules around him and must wear a special suit
- 5 small, small, magnified, large, mechanical, small, large, pressure, equally

- 6 a the same b The wider tin has a greater weight of liquid to support but the downward force from the water being pulled by gravity is spread over a larger area c Salt water is denser than fresh water so the pressure reading would be higher
- 7 a 100 N per cm² (1 000 000 N per m²) b 2000 N
- 8 a no b There are only very small spaces between the particles, so liquids cannot be squashed c yes, quite a bit d Particle numbers will stay the same but the density will increase
- 9 Make the input syringe larger
- 10 a Because water pressure increases with depth, dam walls must be thicker at the base or curved to increase the area and spread the load b The pressure will be greatest at the deepest point, so the water rushes through the turbines very forcefully

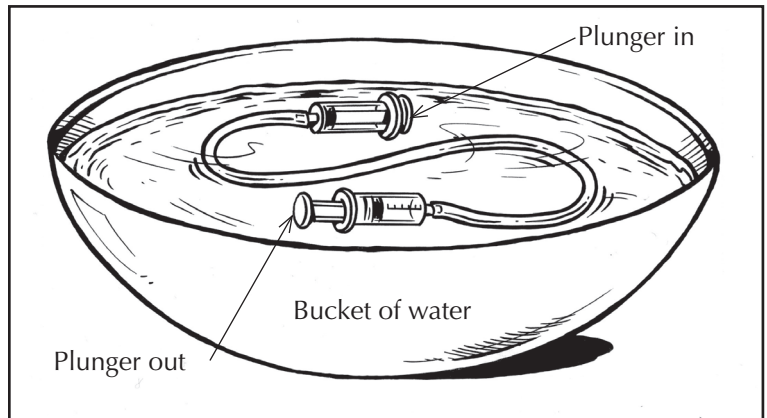
★ Give us a clue – mouse manoeuvres

What you need

Different-sized syringes (1 mL, 10 mL, 20 mL), a few pieces of flexible tubing, a bucket or bowl of water, sheet of pink cardboard, scissors, stapler, sticky tape, empty tissue box, marker pen

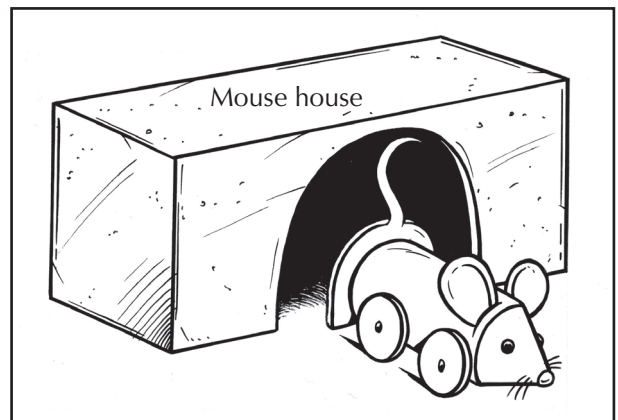
What to do

- 1 **Using** the cardboard, design and construct a small mouse that will fit into the tissue-box mouse house.
- 2 **Fill** one syringe and a piece of tubing with water; fully submerge them in the bucket or bowl of water so that no air is present.
- 3 **Keeping** the syringe and tubing in the water, take a second syringe with the plunger fully compressed and connect it to the tubing under water.
- 4 **Connect** the first syringe, which is full of water and has a fully extended plunger, to the other end of the tube.
- 5 **Remove** the hydraulic machine you have made from the bucket of water, and press the extended plunger all the way in. Experiment with different-sized syringes and different lengths of tubing.
- 6 **Design and build** a hydraulic machine that will move the mouse both into and out of its house.



Now answer True or False to these statements:

- A small plunger moves further but less forcefully.
- A large plunger moves a shorter distance but more forcefully.
- Because a liquid cannot be squeezed, it can be used to transmit forces through pipes.
- Hydraulic machines make things move by putting liquids under pressure.



What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Because liquids cannot be compressed, a change in pressure at any point in a closed system is transferred equally to all points in that system and acts in all directions.

Sticky tape can be used to fix syringes in place and syringes can be hidden inside boxes and tubes.

★ The challenge

Design and build a safe machine combining electrical and hydraulic systems to remotely switch a spa bath or jacuzzi on and off.

Materials

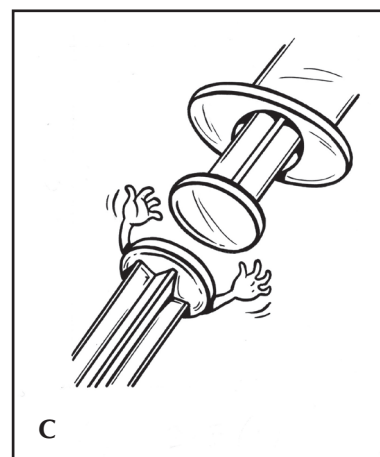
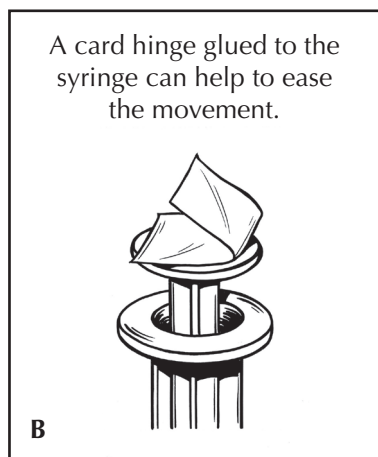
6 volt battery, 1 m of electrical wire, sheet of cardboard, 10 cm of aluminium foil, scissors, masking tape, 4 syringes (2 × 5 mL, 2 × 20 mL), buzzer, 0.5 m flexible tubing, bucket of water

SAFETY!

⚡ Remember that electricity and water are dangerous if mixed.

Instructions

- 1 Use any of the equipment provided that helps you achieve this challenge.
- 2 Instead of connecting this machine to a real spa bath or jacuzzi, use the buzzer provided. If this works, it proves your machine would work properly in real life.
- 3 You may **test** the electrical circuit on its own, but the hydraulic system must not be trialled before the official test.
- 4 The whole device will be tested by the judge.
- 5 You can use the diagrams here for extra clues.



Clues!

Pressure is transmitted equally throughout a fluid and acts in all directions.

Liquid under pressure can be used to move things.

Hydraulics can be useful in electrical work.

Electricity involves a flow of electrons.

A **circuit** is a continuous path of conductors through which electricity travels. A **switch** is like a bridge for electrons; when you lift the bridge out of the circuit, electrons can no longer flow.

★ So what's going on?

Machines

Machines change energy from one form to another. The three basic machines on which all other machines are based are the inclined plane, the lever and the wheel or pulley. Machines make the work we do in our daily lives easier – they give us a **mechanical advantage**. They do this by multiplying forces, changing speeds and changing the direction of forces.

Hydraulics is a Greek word; 'hudor' means water and 'aulos' means pipe. **Hydraulics** use a liquid in a sealed container to transfer a force instantly from one place to another. By manipulating the area over which the force acts, for example, different-sized pistons in a hydraulic jack or press, a mechanical advantage can be achieved.

Hydraulic machines use liquids rather than levers and wheels to make things move. The basis of hydraulics was worked out in about 1650 by the French physicist Pascal, who was studying fluid pressure (gases and liquids).

Pascal's Law: Pressure in a liquid acts equally in all directions, and changes in pressure are transmitted instantly.

The pressure of a liquid in a sealed system is transmitted equally in all directions. For example, when a liquid such as water or oil fills a closed container, the application of a force (pressure) at any point will be transferred to all surfaces of the container.

Pressure in a liquid

- **Pressure depends on the height of the column of liquid.** The force of gravity tries to pull a liquid downwards with a force of 10 newtons per kilogram, which causes pressure on the container and pressure on any object put into the liquid. The pressure on the bottom of a submerged object is greater than that on the top of the object. Pressure depends only on the height of the liquid column – it is not related to the width.
- **Pressure depends on the density of the liquid.** Density is the mass or number of atoms or molecules in a certain volume of a substance. The density of water is 1 g/mL. Petrol is less dense than water, so if a lake was full of petrol instead of water, the pressure at all points in the lake would be less. The opposite would be true for salt water, which is denser than both petrol and water. A hydrometer can be used to measure the density of liquids.

So pressure depends on both the depth and the density of a liquid.

Liquids

Liquids have important properties that make hydraulic machines possible. Liquids maintain their volume but take on the shape of their container. The atoms of a liquid are close together, vibrate slowly and can slide over each other slowly. Liquids can be 'trapped' in a container but they cannot be squashed. Because they cannot be squashed into a smaller volume, pressure changes are felt immediately throughout the liquid. Gases, on the other hand, can be squashed.

Doing the sums

Paths followed by the jets of water

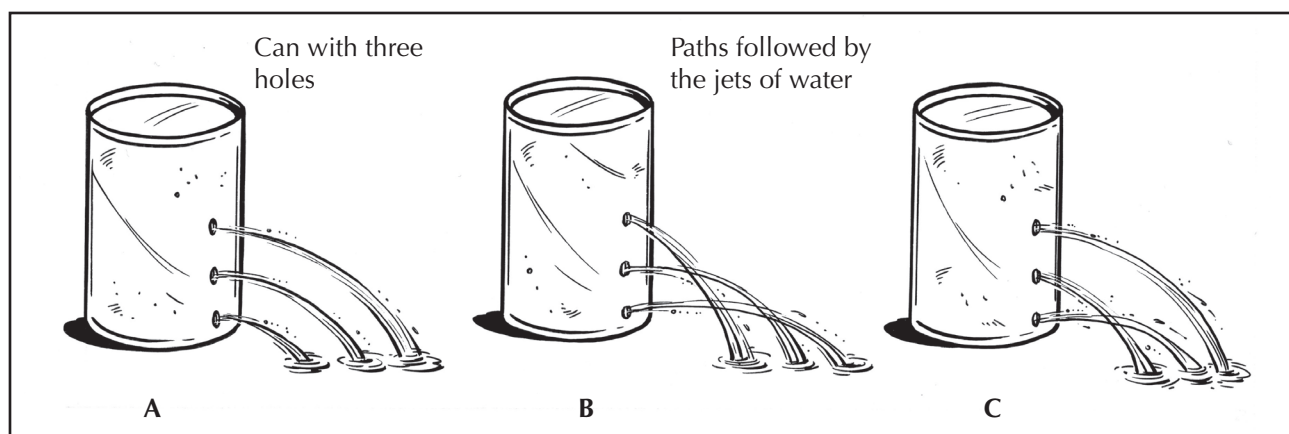
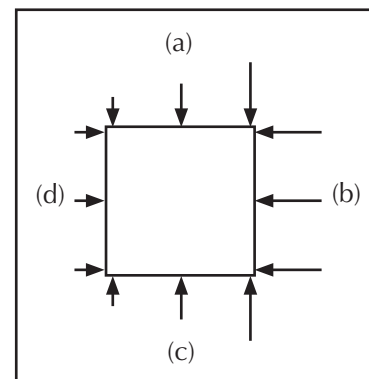
Pressure in a liquid is its 'power to push'. This pressure is affected by the size of the force and the **area** over which the force acts. A force over a small area gives a large pressure.

$$\text{Pressure (newtons per square metre)} = \frac{\text{force (newtons)}}{\text{area (square metres)}}$$

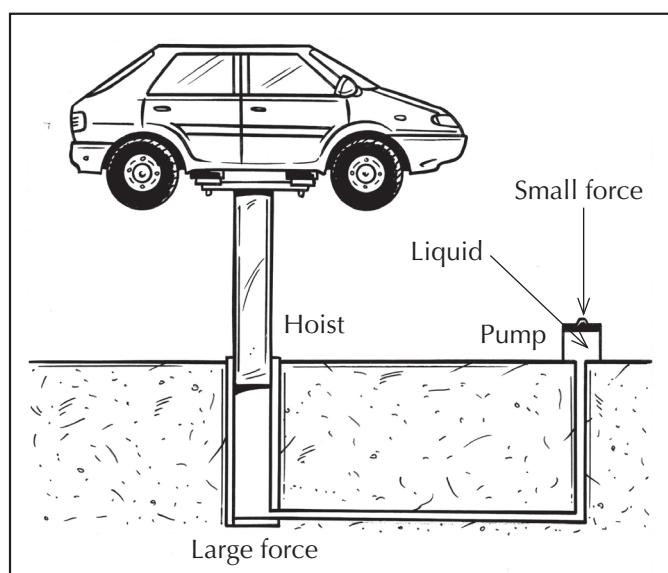
Pressure is measured in kilopascals (kPa). One pascal equals one newton per square metre.

★ Figure it out!

- 1 **a** Name the French physicist who studied fluid pressure in both gases and liquids.
b State the law that explains his findings.
c What two main factors affect pressure in a liquid?
- 2 The object in the diagram is submerged in a liquid. The lengths of the arrows indicate the magnitude of the force acting on the surfaces of the object by the liquid. Which surface of the object is closest to the surface of the liquid?
- 3 **a** List three basic machines on which more complicated machines depend.
b How are hydraulic machines different from these basic ones?
c What two properties of liquids make hydraulic machines possible?
- 4 Study the following diagrams and answer the questions.



- a** Which of the diagrams correctly shows what paths three spurts of water will take from a tin can that is constantly kept full of water by a running tap?
- b** What will gradually happen to each path of water when the tap is turned off and the water drains? Explain your answer.
- c** Has depth got anything to do with pressure in liquids?
- d** Does diagram B support your answer to (c)? If so, how?
- e** What do you think might happen to a scuba diver who goes too deep?
- 5 Study this diagram of the hydraulic press. Fill in the words that are missing from the explanation of how it works (on the next page).



In a hydraulic press, a _____ force applied to a _____ piston is _____ as it is transferred to a second piston with a _____ surface area. This machine delivers a _____ advantage by magnifying a _____ force into a _____ one. Pascal's Law supports this because it states that when _____ is applied to an enclosed liquid, it spreads _____ to all parts of the container.

- 6 Some students were investigating Pascal's Law. They found two tins of the same depth but different diameters and filled them with fresh water. Then they measured the water pressure at the bottom of each tin with a hydrometer.

a Were the pressure readings for both tins the same or different?

b Explain your answer to (a).

c Would the readings be greater or smaller if the tins were filled with salt water instead of fresh water? Why?

- 7 In a hydraulic brake, a force of 500 N is applied to a piston with an area of 5 cm².

a What pressure is transmitted through the liquid?

b If the other piston has an area of 20 cm², what force is exerted on it?

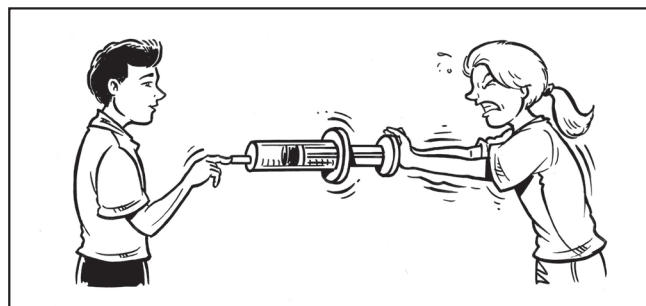
- 8 The two students in the diagram are investigating some properties of liquids and gases.

a If the syringe was filled with milk, could they push the plunger in at all?

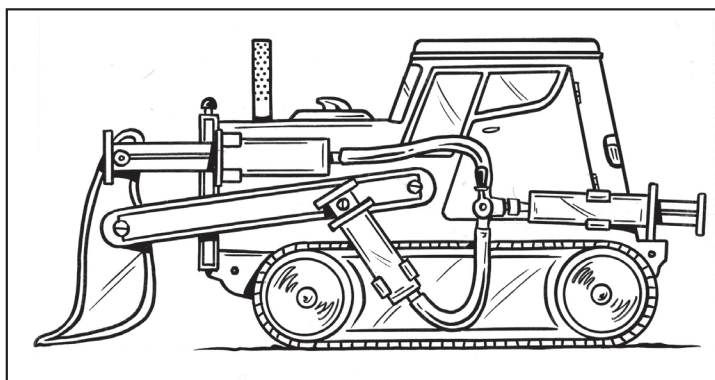
b Explain your answer to (a).

c If the syringe was full of air, could they push the plunger in at all?

d If air can be squashed, what would happen to the number of particles in the syringe, and what will happen to the density of the air?



- 9 Study the diagram of the earthmover. The input syringe cannot fully extend the two output syringes at the same time. Can you think of a way to overcome this problem?



10 Extend your understanding

Dam walls are built to collect and hold vast quantities of water for storage and to generate hydroelectricity. Hydroelectric power stations use moving water to turn turbines and generate electricity. They are often built beneath big dams with deep reservoirs of water.

a Why are dam walls either curved or much thicker at the base than at the top?

b Why are the turbines located at the base of the dam wall?

★ Topple testing

Make sure the dominoes are set up in straight lines. A strip of paper with 1 cm intervals marked on it will make accurate spacing easier. Draw a class graph

showing average speed (y-axis) versus spacing in cm (x-axis), then predict what spacing would allow the dominoes to topple at maximum average speed.

★ The challenge

Dominoes are not expensive, especially if purchased from a two-dollar shop (or similar). Each team will need 100 dominoes. Make sure they are all the same brand, as lengths can vary slightly. This time the dominoes don't need to be set up in straight lines – any patterns and spacings are fine, as long as they don't touch. When the dominoes are close together they will topple slowly, because the speed the dominoes

reach as they fall over is less. When the dominoes are far apart the speed is also slow because it takes them longer to touch the next domino. Dominoes standing on their short sides have a higher centre of gravity and therefore topple more easily.

LINKS TO THE EVERYDAY – chain reactions, energy transfers, nuclear reactions, explosions

★ Curiosity trap

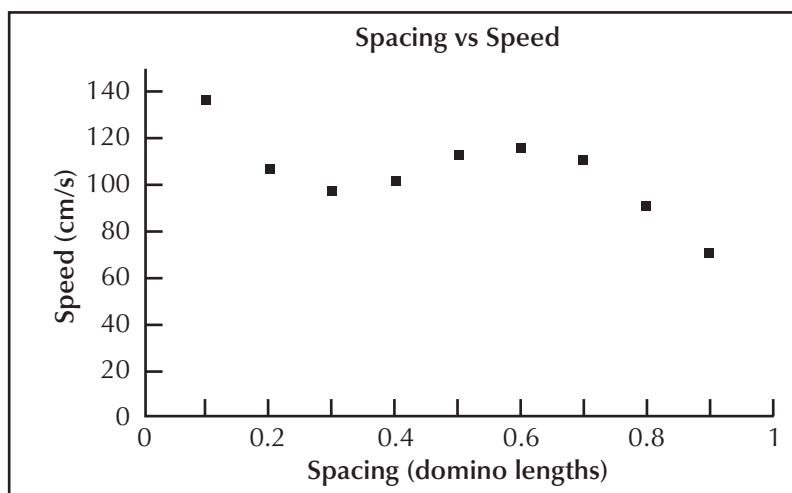
Take 100 dominoes and set them up in a straight row with 2 cm spacing. Push the first domino in the row and use a stopwatch to measure how long they take to topple. This is also a measure of how long it takes for the energy to be transferred.

Have students calculate how many dominoes fall in one second. See if students can tell whether the number of dominoes being knocked over each second increases, decreases, or remains the same throughout.

★ Answers

- 1 speed, dividing, time, average
- 2 a 600 km/hour b 10 m/sec c 160 km/hour
- 3 Each domino is not exactly equally spaced, all dominoes may not be perfectly uniform, and the surface over which the dominoes are placed may not be uniform
- 4 The energy transferred to the last domino is transferred to the floor as it topples over. Some of the energy applied to the first domino will be lost along the way as sound and heat energy
- 5 a 65.8 seconds b speed = distance moved divided by time interval c average speed = 2 m divided by 65.8 seconds = 0.03 m/s d 166.7 seconds (2.8 minutes) e This allowed the car to reach a constant speed
- 6 a the motorist b 10 m/sec c 20 m/s d not unless the speeding motorist slows down or the police car speeds up
- 7 a A chain reaction is a series of similar events that are all related b more c the random pattern d One student sneezes on several students, transferring germs. These students then sneeze on lots more students and a few teachers, transferring more germs. Before you know it there are many students and teachers away from school sick e The engine of a car converts fossil fuel to heat and mechanical energy that makes the car move and releases gases and particles into the atmosphere. These pollutants are related to environmental problems such as acid rain and global warming

- 8 a total seconds (75) divided by 5 = 15 seconds b 30 drops divided by the average time for a 30-drop trial (15 seconds) = 2 drops/sec c At the rate of 2 drops/sec, the race took 45 seconds to run. Speed equals distance divided by time: 180 m divided by 45 sec = 4 m/s
- 9 a uncontrolled b model 2 (controlled chain reaction) c uncontrolled
- 10 a refer to the graph below (spacing versus speed) b If a row 100 cm (1 m) long takes 1 second to fall, then a row 60 times as long (60 m) would take 1 minute to fall c 60 m/minute or 100 cm/sec d When the dominoes are close together the speed is slower because the speed of dominoes falling over is less. When the dominoes are far apart the speed is slower because it takes longer to touch the next domino



★ Give us a clue – topple testing

What you need

50 dominoes, 1 m ruler or tape measure, stopwatch,
large benchtop or floor space



What to do

- 1 **Take** 50 dominoes and stand them up on their short side, in a straight line so that the space between each domino is 2 cm. The dominoes must not be touching.
- 2 **Measure** the length of the row of 50 dominoes in centimetres using the ruler.
- 3 **Start** them toppling by pushing the first domino over and measure the time it takes for all of the dominoes to topple. You may need to practice this a couple of times.
- 4 **Record** all your measurements in the data table below and work out the average speed of each topple.
- 5 **Change** the space between the dominoes to 0.5 cm and record the row length as well as the topple time.
- 6 Now **you decide** on two more ways to arrange the dominoes in a straight line and then measure the row length and topple time.
- 7 **Repeat** these tests with the dominoes on their long side.

Data table

DOMINO SPACING	DOMINO ROW LENGTH (CM)	TOPPLE TIME (SEC)	AVERAGE SPEED (CM/SEC)
2.0 cm			
0.5 cm			
Own choice			
Own choice			

Now answer these questions:

- Is there any relationship between the average spacing distance of the dominoes and the average speed at which they fall?
- Does the length of the side on which the dominoes stand affect the speed of the topple?

What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

The average speed of the dominoes falling is the total distance of the domino row divided by the time it took for the dominoes to topple.

★ The challenge

Set up 100 dominoes in such a way that they take the longest possible time to topple when set off by a single domino toppling.

Materials

100 dominoes, ruler, stopwatch, large smooth floor space

Instructions

- 1 You have 15 minutes to work out how to **set up** a pattern using 100 dominoes, spaced in such a way that they take the longest possible time to topple. You may test your ideas as many times as you like.
- 2 You may **use** any size of space between the dominoes as long as it is more than the thickness of one domino. The dominoes must not touch in any way.
- 3 **When** you have worked out the best plan, tell the judge. You will then have 10 minutes to set up the 100 dominoes again ready for testing.
- 4 **The judge** will start the topple by gently tipping the first domino onto the second one. The judge will time the entire topple. If the topple stops before all the dominoes have fallen over, the final time will be recorded at the point when the topple stops.

Clues!

*Speed is the rate at which an object is moving.
It combines information about how far and for
how long an object travels.*

*A pattern is a design, a shape or a model.
Science involves understanding patterns.*

★ So what's going on?

Energy transfer

Energy makes things happen, but only if it is transferred. Energy can be transferred from one object to another and from one particle to another in solids, liquids and gases. Energy can never be used up and it can never be created – except in nuclear reactions.

Moving along

A central property of all moving things is speed. Speed is the rate at which distance is covered by a moving object. The direction in which the object is moving is of no importance. By 'rate' we mean how much, or how many of something in a certain amount of time. For example, how many kilometres travelled in one hour or how many raindrops hitting a roof in one minute. The preferred unit for describing speed is metres per second. If we measure the speed of sound or the speed of light, we measure the rate at which this energy moves. We cannot see energy moving but we can see and measure what it does as it moves. By using a row of falling dominoes we can measure the speed of energy transfer.

Speedy calculations

To work out the speed of something, you need to know how far it travels (distance) and the time it takes to travel this distance.

$$\text{Speed (m/sec)} = \frac{\text{distance travelled (metres)}}{\text{time taken (seconds)}}$$

If an object's speed does not change from the beginning of its journey to the end then it is moving at a **constant speed**. Most things, however, do not move at the same speed all the time. For example moving cars, bicycles and people all speed up and slow down. So the speeds you can work out using this formula are actually their **average speeds**.

Looking for patterns

Science involves solving problems, and one of the tools that can help you do this is to look for patterns. This involves looking for relationships between things. Being able to identify and work with patterns is an important scientific skill.

The chain of reactions

When one event triggers another similar event, which in turn causes more of the same or similar events, a chain reaction is taking place. Sometimes this can release large amounts of energy. Toppling dominoes is one example of a chain reaction. Another example is nuclear fission: when a neutron strikes a uranium atom, the atom's nucleus splits, producing two or three free neutrons. When these neutrons strike other atoms, the chain reaction called fission occurs. Controlled nuclear fission chain reactions are used in nuclear power plants. The rate of the fission chain reaction is managed carefully so that the heat energy produced is released at a steady rate. An uncontrolled chain reaction would cause an explosion. Uncontrolled fission reactions produce enormous amounts of heat energy very quickly, such as in atomic bombs.

The rate of reaction in a chain reaction, if allowed to go on unchecked, gets faster and faster in a very short space of time. The amount of time it takes for 100 dominoes to fall down shows how a large number of nuclear collisions in a short space of time can sustain a nuclear chain reaction.

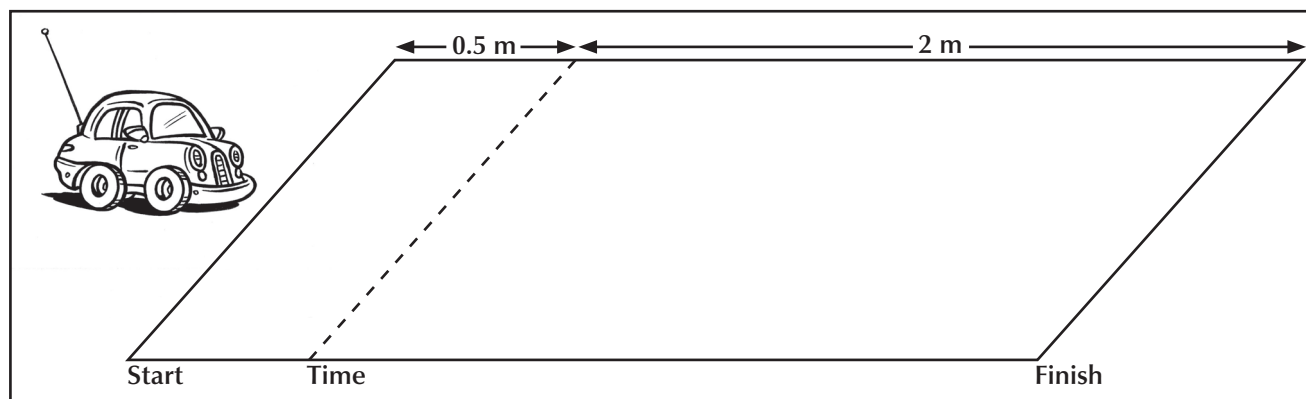
★ Figure it out!

- 1 Read the information below and fill in the missing words.
The distance something moves in a certain time is called its _____. You can work out speed by _____ distance by _____ taken. Most things don't move at the same speed all the time so the speed we work out is called _____ speed.
- 2 Work out the average speeds for the following moving objects:
 - a An aeroplane travels 3000 kilometres in 5 hours
 - b An athlete runs 100 metres in 10 seconds
 - c A train travels 240 kilometres in 1.5 hours
- 3 The speed at which dominoes fall is probably not constant. Suggest possible reasons for this.
- 4 Imagine placing ten 20-cent coins in a straight row on a table. Then you slide a 50-cent coin across the table and knock the first 20-cent coin in the line. You would find that the last 20-cent coin in the line will shoot out on its own as the energy is transferred from one coin to another along the line and then causes this last coin to move.

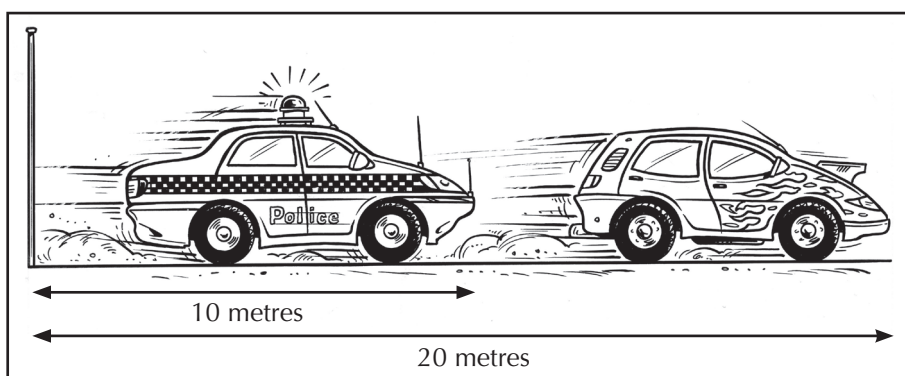
With the toppling dominoes, where does the energy go once it has been transferred to the last domino?
- 5 The speed of an object is determined by the amount of time it takes to move over a particular distance. The diagram below shows a 2.5 m race track with 'start', 'time' and 'finish' lines marked. The toy car is switched on at the 'start' line. Timing begins at the 'time' line and ends at the 'finish' line. Study the following data table and answer the questions.

Data table

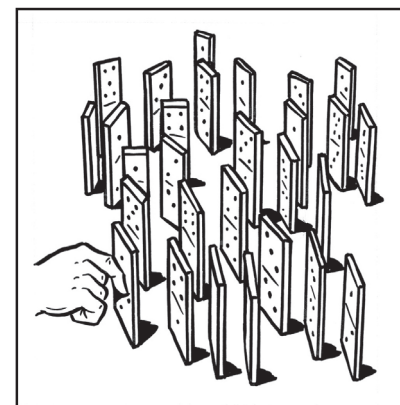
	TRIAL 1	TRIAL 2	TRIAL 3
Time to travel 2 m (seconds)	66.2	59.8	71.4



- Using the interval times from the three trials, calculate the average travel time over the 2 m.
 - State the mathematical relationship between speed and distance.
 - Calculate the average speed of the toy car.
 - How long will the car take to travel 5 m?
 - What was the purpose of letting the car run 0.5 m before starting to time it?
- 6 Look at the drawing of the police car and the speeding motorist. They both passed the lamp post at the same time. The drawing shows how far the police car and the motorist went in the next second.



- Which is moving the fastest, the police car or the speeding motorist?
 - How fast is the police car going in m/s?
 - How fast is the speeding motorist going in m/s?
 - At these rates, will the police car ever catch the motorist?
- 7 Dominoes are sometimes used to demonstrate a chain reaction. The diagram shows a random pattern of dominoes such that each domino will hit several other dominoes at the same time.



- What is a chain reaction?
- Would the number of dominoes being knocked over per unit time in the random pattern be more, less or the same as when dominoes are set up in a straight line?
- Which approach, the random or straight pattern, takes the shorter time to knock over all the dominoes?
- How is a chain reaction similar to the flu spreading quickly throughout a school?
- Solving environmental problems is difficult because all parts of the natural world are connected. How does driving a car start a chain reaction that helps pollute our atmosphere?

- 8 A group of students designed a clock, using a water bottle and an aluminium pie dish. The water flow from the bottle was adjusted to a steady, regular drip. They did five trials, measuring the time it took for 30 drips.

Data table

TRIAL	SECONDS FOR 30 DROPS TO FALL
1	13
2	18
3	17
4	12
5	15

- a Use the data for the five trials to calculate the average number of seconds required for 30 drops to fall into the dish.
- b Use the answer for (a) to calculate the average number of drops per second.
- c The students used this clock to time a 180 m sprint. They counted 90 drops from the start to the end of the race. Calculate the winning athlete's speed in m/sec.
- 9 Students set up two models of a chain reaction.
- For model 1, they arranged dominoes in a pattern so that each domino hit two other dominoes, which then in turn hit two other dominoes. The more dominoes they used, the more dramatic the effect.
- For model 2, they set up the dominoes so that most dominoes hit only one domino or no dominoes at all, with only one or two dominoes able to hit two others as they fell.
- a Is model 1 an example of a controlled fission chain reaction or an uncontrolled fission chain reaction?
- b Are nuclear power plants based on model 1 or model 2?
- c Is an explosion a controlled or uncontrolled chain reaction?

10 Extend your understanding

Some students who were studying moving objects recorded the following data on falling dominoes.

TEST NUMBER	SPACE BETWEEN DOMINOS (DOMINO LENGTHS)	SPEED (CM/S)
1	0.1	135
2	0.2	105
3	0.3	95
4	0.4	100
5	0.5	110
6	0.6	115
7	0.7	105
8	0.8	85
9	0.9	70

- a Draw a graph of this data where y = speed (cm/s) and x = spacing (domino lengths).
- b Study the relationships recorded in the graph and predict how long a row of dominoes would have to be to take 1 minute to fall.
- c At what average speed would this row be toppling?
- d Explain the shape of the graph.

★ Bubble-ology

Ensure you use a good quality dishwashing detergent. Morning Fresh (no lemon) always gives very good results. For a basic bubble mix, combine 2L dishwashing liquid with 6L water, then add about 50mL of glycerine (from a pharmacy). This mixture works particularly well if it can be made up a couple of weeks early and allowed to stand. If this is not possible it will still be fine. The mix can be stored for long periods of time at room temperature, kept covered, and can be used many times over. If you live in an area with hard water then use distilled water for a better bubble mixture.

Detergent reduces the surface tension of the water to about one third of what it normally is and allows just the right amount of attraction for bubbles to be blown.

However, water tends to evaporate quickly, and once it does the bubbles will burst. Adding glycerine delays evaporation. Keep in mind that as long as surfaces, hands and equipment are wet with bubble mix, you can touch soap films without popping them. Bubbles can be made by blowing air into a skin or by waving a frame around to trap air inside.

It is difficult to blow bubbles outside if there is any wind at all, so use lots of newspaper or dropsheets to cover benches and parts of the floor. Mop up any spills immediately and avoid areas with carpet. Students should roll up their sleeves, wear rubber-soled shoes and remove any jewellery, especially watches on their arms. Detergent can irritate eyes and cause allergic skin reactions, so rinse with plenty of fresh water.

★ The challenge

Have a bucket of clean water for each group. Tie-wire can be purchased from hardware stores and craft wire from craft shops. Strong scissors will cut wire if wire cutters are not available.

When a bubble bursts, it leaves a ring of small droplets. By measuring the diameter of the ring, bubble sizes can be compared. Burst the bubbles on

coloured cardboard so that the rings are easy to see and provide a record of each test.

LINKS TO THE EVERYDAY – raindrops, bubbles in shampoo and soaps, insects that can walk on water, spirit levels, eye droppers, water droplets on a spider's web, bubble bath, packaging materials, foams, soft drinks, breads and cakes.

★ Curiosity trap

A good analogy to explain a bubble is a balloon. A balloon is a stretchy skin inside which we can trap air; bubbles are the same. Develop this concept by blowing two bubbles with a medium-sized funnel, one from the wide end and one from the narrow end. Bubbles can be different sizes depending on the tool used, but they are mostly spherical. However, with the right tool we can make a square bubble. Form a

cube shape with a handle using pipe cleaners.

Dip the cube into deep bubble mix and study the pattern formed. Then carefully dip the bottom face of the frame into the bubble mix a second time. When you gently lift the frame out, you should see a cube-shaped bubble in the centre. Gently pop the films on each side of the cube frame one at a time until a square bubble remains.

★ Answers

- 1 **a** a thin film of soap and water filled with air **b** surface tension **c** The tendency for the surface of a liquid to act as if it was covered by a skin due to the force of attraction between the surface molecules **d** yes – but it may not be as strong as in water
- 2 **a** Bubbles are almost always spherical **b** Water molecules pull as close as possible to each other **c** air movement **d** A sphere is the shape with the least surface area for a particular volume
- 3 The skin on the water is usually broken by your hands as they enter the water before your body, but when you do a belly flop the skin does not break to let you through
- 4 The powder floats on the skin until the detergent reduces the surface tension. As the surface water 'relaxes', the powder is pushed outwards and sinks
- 5 great, soap, detergent, reduces, stretchier
- 6 **a** the detergent component **b** 20% **c** no **d** No! Because a dishwasher is a closed system, the bubbles would build up and up inside the machine, making a

- real mess. Because we can't see inside the machine when it is operating, we won't know that there are no bubbles
- 7 The molecules of water stick together and are surrounded by a skin which is like a small bag. When a droplet holds so many molecules that the skin is stretched to its limit, it will break away from the tap and fall
- 8 The detergent reduces the surface tension at the opening, so the skin pulls the bread tag more strongly at the front than at the back and the speed boat is pulled forward
- 9 You could pour a weak solution of detergent onto the surface to destroy the surface tension. Then the female mosquito would have trouble landing on the surface to lay her eggs and the wrigglers could not 'hook' on
- 10 **a** The film over the mouth of the funnel will be flat **b** The film will curve up into the funnel when the finger is removed **c** The air pressure on the flat film will cause it to curve into the funnel

★ Give us a clue – bubble-ology

What you need

20 cm of heavy craft wire, scissors, a few straws, pipe cleaners, 0.5 m of knitting wool, 2 L ice-cream container of bubble mix

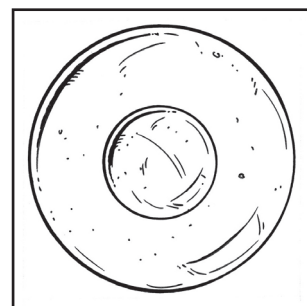
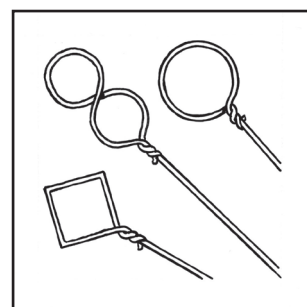
*Bubble buster rules: Every bubble is sacred!
Move around the room carefully and always avoid bursting the bubbles.*

SAFETY!

- ☼ Avoid getting bubble mix in your eyes – it can sting. If you do get some in your eye, wash it out with plenty of water.
- ☼ Bubble mix can be slippery, so clean up spills and drips as they happen.
- ☼ Walk around the room carefully: do not run.
- ☼ Wire pieces may have sharp ends, so take care.

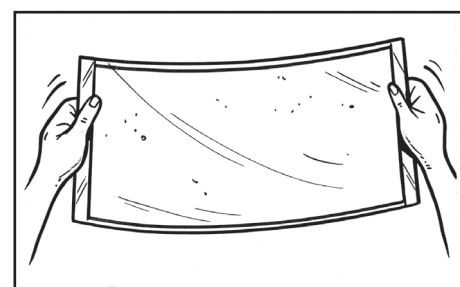
What to do

- 1 **Make a hoop** out of wire and blow some bubbles. Try to answer these questions:
 - How large a bubble can you blow?
 - How small a bubble can you blow?
 - How long can you make a bubble last?
 - How high can you make a bubble float?
- 2 **Bend the wire** into different geometric shapes, like diamonds and figure of eights. What happens when you blow bubbles through these shapes?
- 3 **Wet your finger** with bubble mix and see if you can push it through a bubble without bursting it. Try this with a straw and see if you can blow a small bubble inside the first bubble.
- 4 **Make a bubble frame.** Cut a straw into two shorter straws and run the knitting wool through the two straws. Knot the string and pull the knot inside one of the straws. Put the frame into the bubble mix and lift it out carefully so that there is a soap film 'window' stretched across the rectangle.



Experiment with the frame and answer these questions:

- When you hold the rectangle to the light, can you see rainbow colours?
- If you twist the frame around, can you make interesting bubble shapes?
- Can you push your hand through the 'window' film without breaking it?
- Can you make a super-bubble by holding the frame at arms' length and pulling it towards you? (To close the bubble off, try bringing the two straws together until they touch as you swing the frame upwards.)
- Can you blow a bubble using no tools, only your hands?



What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

★ The challenge

Blow the biggest spherical bubble you can with the equipment provided, then measure its size in centimetres.

Materials

2 L of bubble mix, 15 cm of tie wire, wire cutters or strong scissors, ruler, large sheet of coloured cardboard, 1 m of knitting wool, medium-sized plastic funnel, shallow plastic containers, 6 straws, paper towels

★ Instructions

You have 15 minutes.

- 1 **Use** any of the equipment provided as bubble blowers, or use the equipment provided to construct bubble blowers of your own design.
- 2 Carry out however many **tests** you like, with as many bubble blowers as necessary.
- 3 Develop a **technique** that allows you to successfully blow large bubbles.
- 4 **Decide** on the bubble blower that will enable you to produce the biggest spherical bubble possible.
- 5 **Work out** a way to accurately measure the size of the bubble.
- 6 The judge will watch you blow the challenge bubble and accurately **measure** the size of this bubble. You are allowed only two attempts at blowing this bubble; the larger of the two will be recorded for your official result.

Clues!

Bubbles burst when they dry out, and they will burst if they touch a dry surface.

Blowing is not the only way to trap air inside a skin.

It is much harder to make bubbles from a stirred-up bubble mix with suds on top, so skim off the suds regularly.

★ So what's going on?

Stretchy skins

Scientists define bubbles as gas encapsulated by a substance that is, or was, a liquid. Most people only think of soap bubbles, but other kinds of bubbles play an important part in everyday life. For example bubble wrap, soft drinks, cakes and bread. Bubbles are almost always spherical, because the water molecules pull as close as possible to each other. However, a very large bubble can become distorted by air movement. A sphere is the shape with the least surface area for a particular volume.

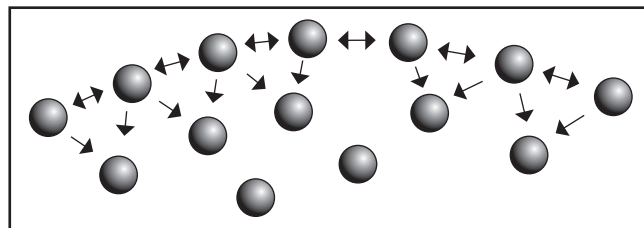
A bubble film can contain as many as 150 different layers, but because bubble films are so thin, it would take thousands of bubble walls to equal the thickness of a human hair! When light passes through a bubble film, it is reflected from the top and bottom layers of the film, producing all the colours of a rainbow. We see a scrambled combination of colours because the layers are uneven in thickness.

Physicist Eiffel Plasterer blew a bubble that lasted 340 days, and Richard Faverty created bubbles 4 metres in diameter and bubble tubes 10 metres long.

Adding soap or detergent to water makes it 'stretchier' so it can form thin, elastic films, ideal for making bubbles. This is because the soap or detergent reduces the water's surface tension by about two-thirds. The surface tension of ordinary water is too great to allow bubbles to form.

Surface tension

The molecules of liquids, including water, tend to pull in towards each other equally and in all directions. We say that water sticks to itself. Water molecules that are surrounded by other water molecules are attracted in all directions, but those on the surface of a body of water are being pulled from below and from the sides by stronger forces than from the top (air), so the water pulls its surface close over itself. It's as if the molecules stick together, forming a thin skin that is tight, fairly strong and a little elastic. Drops of water have a 'skin' all around them, like a small bag, but bodies of water only have a skin on the surface. This skin is flexible enough to stretch a little and strong enough to support the weight of a small insect. Soap and detergent molecules reduce the tension between the molecules of water, but bubbles are also held together through surface tension.



Suds – the real truth!

The advertisers of soaps and detergents want us to believe that lots of suds means really good cleaning power, but really suds have very little to do with it. These products are meant to remove dirt from solid surfaces. Detergents are made from either natural fats and oils or petrochemicals. These substances make up about 20% of the soap or detergent. The other 80% is made up of things like silicates and phosphates, to improve the cleaning action. Other additives may alter the acidity and alkalinity of the mixture or the brightness and colour of the cleaner. The detergents do the real work, not the bubbles floating on top. Because consumers think bubbles mean good cleaning power, manufacturers make sure their products produce bubbles when used, so that people believe they work.

The last word: If you take a huge breath, you become a human bubble.

★ Figure it out!

- 1
 - a What is a bubble?
 - b What holds bubbles together?
 - c Explain the phenomenon in your answer to (b) and how it works.
 - d Do liquids other than water have surface tension?
- 2
 - a What shape are most bubbles?
 - b Why are they this shape?
 - c What force may cause a very large bubble to distort?
 - d What is the mathematical relationship between the surface area of a bubble and its volume?
- 3 When you dive into a swimming pool, why does it hurt if you do a 'belly flop' instead of performing a perfect entry into the water?
- 4 A group of students were experimenting with surface tension. They got a bowl of water and sprinkled a thin layer of talcum powder all over the surface. They found that the powder floated on the surface. When they added a drop of detergent to the middle of the water surface, the powder pulled towards the edges of the bowl and then began to sink.

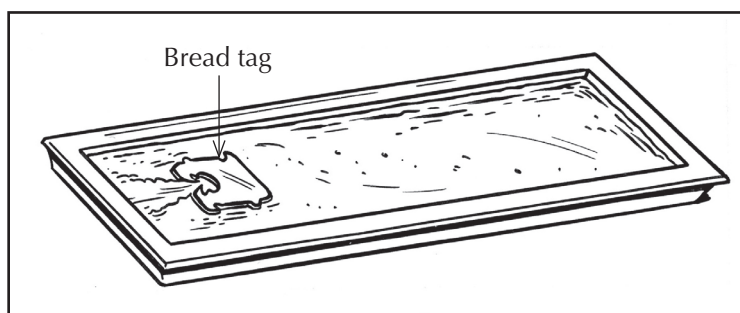
Using your knowledge of surface tension, explain this result.

- 5 Bubble-ology is a very delicate and careful science. Add the missing words to the following information about the study of bubbles.

The surface tension of ordinary water is too _____ to allow bubbles to form. However, adding _____ or _____ to water _____ the surface tension of water to about a third of what it is normally. The water becomes _____ and the mixture can form thin, elastic films – ideal for making bubbles.

- 6 To help you answer the following questions, think back to when you last washed the dishes, had a bath or helped wash a car. Were there lots of bubbles as you were cleaning?
- a** What part of a cleaning product actually removes the dirt from surfaces?
- b** What percentage of cleaning products do these substances make up?
- c** Is the cleaning power of a product improved if it produces lots of bubbles as you clean?
- d** Do you believe that the detergents used in dishwashing machines are manufactured with bubble production in mind? Explain your answer.
- 7 Explain why water dripping from a tap forms a continuous stream of separate, individual droplets.
- 8 A clever student decided to use surface tension to have some fun. She filled a baking tray with water and let it settle. Then she placed a plastic bread tag at one end of the tray, as shown in the diagram. Lastly she dipped a toothpick in detergent and touched the water between the two arms of the bread tag. The bread tag became a speedboat and raced to the other end of the tray.

Explain why the bread tag whizzed to the other end of the tray.



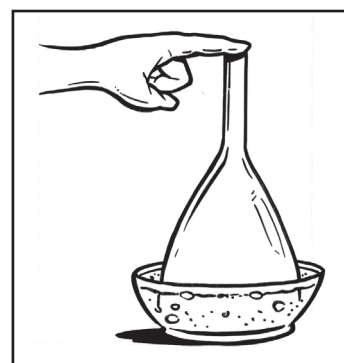
- 9 Many mosquitoes lay their eggs on the surface of water. The female rests on the surface while she lays the eggs. When the eggs hatch into larvae (wigglers) they 'hook' their breathing tubes onto the surface to get air.

What could you do to control such mosquito populations?

10 Extend your understanding

The diagram shows a person holding their finger on the pipe of a funnel and dipping the large end into bubble mix.

- a** When they lift the funnel out of the mix, will the soap film at the mouth of the funnel be flat or rounded?
- b** If the person now removes their finger from the funnel, will the film remain the same or will it change?
- c** Explain your answer to (b).



★ Perfect landings

If safety is a concern when dropping the parachute from a balcony or stairwell, use instead a light cotton cloth the size of a man's handkerchief instead of the

tissue paper; fold up the parachute and toss it high into the air. Students will learn about the effect of air resistance in everyday situations.

★ The challenge

Cheap plastic toy cars about the size of a large coffee mug can be purchased from discount stores. If there are no two-storey buildings in the school, check to see if sports facilities could suffice for at least one level. If possible, the toy vehicle should land on a hard surface such as concrete. More information about collisions can be found in 'Daredevil eggs', in Book 1 of this series.

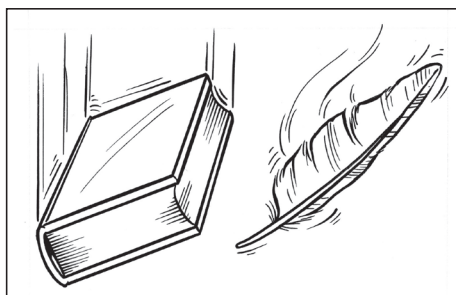
Gravitational force accelerates the toy towards the Earth, so with each passing second it falls faster and

faster. The frictional force of air resistance is also becoming larger and larger. The toy hits the ground with a force equal to the gravitational force pulling on it. The ground pushes up and makes the toy decelerate quickly. In this challenge the toy vehicle does not fall far enough to be able to reach its terminal velocity, so it continues to accelerate towards the ground.

LINKS WITH THE EVERYDAY – skydiving, cargo drops, braking systems for high-speed vehicles such as space shuttles.

★ Curiosity trap

Many people believe that heavy objects fall faster than light objects. Compare the rate of fall of a book and a large feather. Hold the two objects at shoulder height, one in each hand, and drop them. The book will hit the ground first, while the feather slowly wafts down. Now place the feather on top of the book so that it sits completely flat, and drop them both together. The feather drops just as fast as the book.



In the absence of air friction, all objects in a gravitational field fall with the same acceleration. For the book and feather, the relationship between air friction and weight are very different. When the feather is placed on top of the book, the book pushes the air out of the path

of the feather, reducing the force of air resistance on the feather so that they drop together. In a vacuum they would both fall at exactly the same rate.

★ Answers

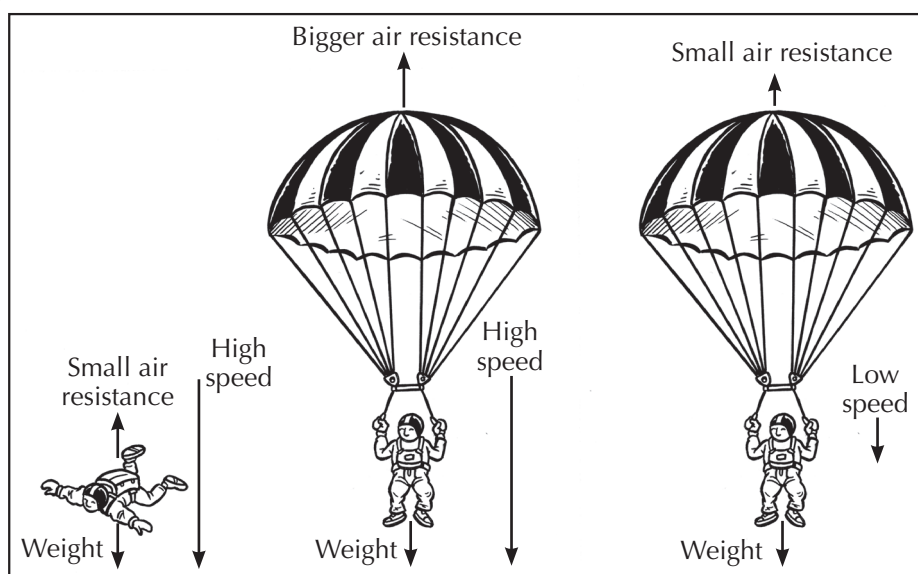
- 1 a air resistance b acceleration c speed d constant speed e velocity
- 2 a Gravity pulls it down and air resistance or drag slows its fall b Without holes the trapped air escapes unevenly from the sides of the parachute and tends to make it fall unevenly c An open parachute canopy is designed to make the air resistance as large as possible, thus slowing the rate of fall
- 3 forces, balanced, steady, gravity, air resistance
- 4 a The pinecone and the winged seed b The larger surface area creates more air resistance, slows their fall and lowers their terminal velocity c shape
- 5 a shuttle: yes, low; parachute: no, high b no – the Moon has no atmosphere to supply drag
- 6 a 180 newtons downwards b Newton's Second Law c When the forces of gravity and drag are balanced d Newton's Second Law
- 7 a 50 m/sec b They would both hit the ground at the same time – in a

vacuum all bodies fall at the same speed c Galileo

8 Refer to the diagram below

9 a 10 m/s b changes c It is accelerating because it is experiencing a constant change in speed

10 a acceleration b terminal velocity c parachute opens d deceleration e terminal velocity



★ Give us a clue – perfect landings

What you need

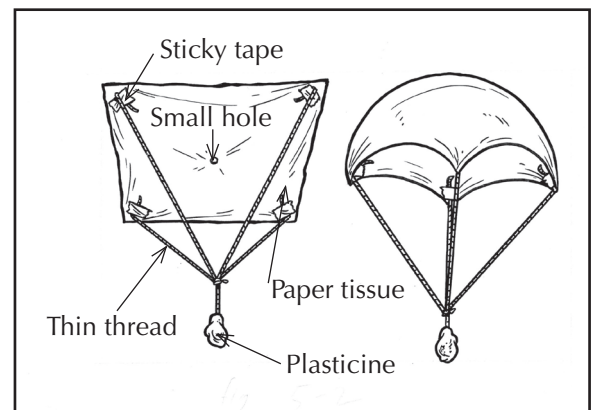
A standard sheet of tissue wrapping paper; scissors; sticky tape; 50 g ball of plasticine; ruler; 4 lengths of thread, each 30 cm long

What to do

- 1 **Cut** a 30 cm square from the tissue paper and carefully tape one end of each piece of thread to each corner of the tissue paper.
- 2 **Bring** the other ends of the threads together and bind them with a small piece of plasticine to form a parachute. Fold the parachute neatly so that it will open out when dropped.
- 3 **Without leaning over railings, drop** the parachute from a high place such as a stairwell or balcony; carefully watch how it behaves as it falls.
- 4 Now **cut** a small hole in the middle of the tissue paper and do the drop test again. Observe the parachute's behaviour as it falls.
- 5 **Modify** this basic design until the parachute will carry the whole ball of plasticine accurately and gently to the ground.

SAFETY!

- ⚠ Breaching safety rules may cause immediate disqualification!
- ⚠ Do not lean out over railings or from windows. Hold the parachute out at arm's length and then drop it.



Now answer these questions:

- Did you notice that the parachute wobbled as it descended?
- What difference did the hole make to the amount that it wobbled as it fell?
- What modifications did you make to your parachute to ensure that it landed accurately and gently?

What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

A parachute is affected by two forces: gravity pulls it down and air resists its movement.

The larger a parachute's surface area, the more air resistance it meets and the slower it falls.

A small hole in the canopy allows a little air to escape as it falls, so its descent is smoother.

★ The challenge

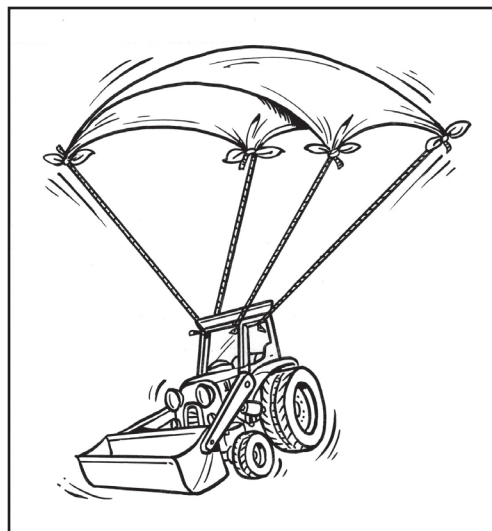
Design a parachute that will deliver a toy vehicle to a child two floors below you, without breaking the toy.

Materials

1 sheet of A4 paper, plastic wrap, 30 cm square piece of lightweight cloth, ball of light string, sticky tape, ruler, scissors, stopwatch, plastic toy vehicle.

Instructions

- 1 Use any of the materials provided to help you design and build the parachute.
- 2 You may **test** your designs by dropping them from as high as you can reach with your outstretched arm. If the toy is damaged in any way before the official test, your team is disqualified.
- 3 The judge will **measure** how long it takes for your final design to reach the ground floor from the second floor. The team whose parachute takes the longest time to fall will score the most points.



Clues!

A parachute is a device that slows an object's fall from a great height, such as from an aeroplane.

An open parachute causes a force (drag) in the opposite direction to gravity and slows the fall.

Since the shape of an object affects its air resistance, the rate of fall can be changed by modifying the object's shape.

The longer deceleration takes at impact, the smaller the force on the object.

★ So what's going on?

Parachutes

A parachute is affected by two forces: gravity pulls it down towards the centre of the Earth while air resists its movement. We use **air resistance** to slow things down. Air resistance is the friction force on something moving through air (drag); it increases as speed increases. **Friction** is a force that acts in the opposite direction to something that is moving or trying to move. Without a parachute, the force of gravity pulls a person to earth too fast. An open parachute is designed to make the air resistance as large as possible. Parachutists use air resistance to slow the speed of their fall, reducing the force at impact and allowing them to land without injury. Forces are measured in newtons (N).

Parachute jumping for sport is called 'skydiving'.

When a skydiver jumps from an aeroplane, the force of gravity (weight) makes him accelerate towards the ground because the forces are unbalanced (the pull of gravity is more than the drag of air resistance). The faster he falls, the greater the air resistance becomes. The air rushing past his ears is absolutely deafening. Eventually the two forces are equal and balanced, so he stops accelerating and falls at a constant speed. This speed is called his **terminal velocity**. When his parachute opens, the force of air resistance increases, so he slows down or decelerates until the two forces are balanced once again. He is now travelling at a new constant speed – his new terminal velocity.

Galileo proved that if we ignore the effect of air resistance, objects will fall at the same constant acceleration no matter how much they weigh. When air resistance is negligible, an object will 'fall freely', for example the skydiver immediately after he leaves the aeroplane.

A word from Newton

The balancing of forces we discussed above is an example of **Newton's First Law**. Because there is no resultant force on the object, it will continue to move at a constant speed once it reaches terminal velocity, in a straight line downwards.

Newton's Second Law says that the amount an object slows down or speeds up varies with the size of the force and the mass of the object. In other words, the larger the force and the lighter the object, the greater the acceleration.

Coming to terms with the terms

The term we use to describe 'how far' and 'in how much time' an object moves (rate of motion) is **speed**. If we add 'in what direction' to this information, the term used is **velocity**. Changes in speed and velocity are measured to work out **acceleration**. In other words, acceleration is the rate at which an object speeds up or slows down. Very few things move at the same speed for long. Nearly everything in the universe is constantly accelerating or decelerating.

Objects falling through the Earth's atmosphere are simultaneously acted upon by gravity and air resistance. As an object accelerates towards Earth, its air resistance (drag) increases. However, the force of gravity remains constant. At a certain speed the upward force of air resistance will equal the downwards force of gravity. Once these opposing forces are equal, the object cannot keep accelerating. It continues to fall at that **constant speed** until it hits the ground.

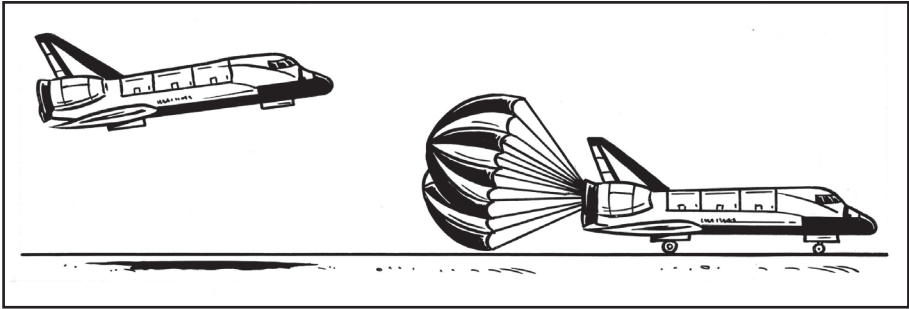
The speed at which the force of air resistance equals the force of gravity is called the **terminal speed** or **terminal velocity**. Each object will fall at its own particular terminal velocity.

The last word: A person with a parachute falls at a rate of about 5 m/sec or 20 kph. Without a parachute the speed would be over 50 m/sec or 175 kph. At either speed, a collision with the earth is dangerous.

★ Figure it out!

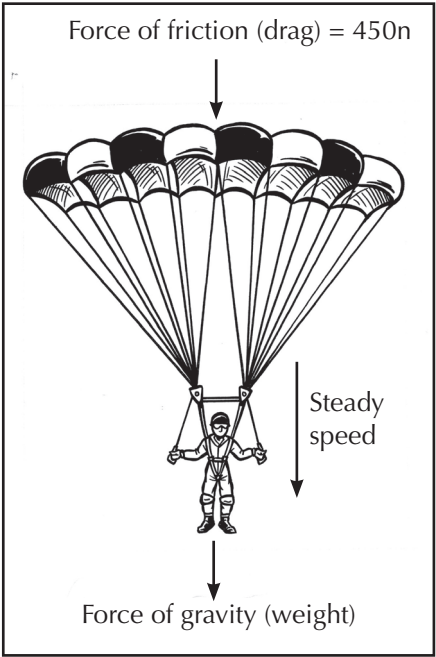
- 1 Provide the correct terms for the following definitions.
 - a A force exerted on a moving object opposite to its direction of motion, due to friction between the object and the air.
 - b The rate at which an object speeds up or slows down.
 - c The rate of motion of an object – how far it travels and how long it takes to travel that distance.
 - d An object moving the same distance within every equal time interval.
 - e Speed in a particular direction.
- 2
 - a Which two forces affect a parachute?
 - b Why do parachute canopies have holes in them?
 - c What is the canopy of a parachute designed to do?
- 3 The following information explains terminal velocity. Fill in the missing words.
 At the terminal velocity the _____ on the object are _____ and it falls at a _____ speed. So the force of _____ (weight) downwards equals the force of _____ upwards.

- 4 The terminal velocity of an object depends on a combination of factors, such as its shape, surface smoothness and composition.
- a Which two of the following would reach slower terminal velocities if dropped from a helicopter: an arrow, a pinecone, a raindrop, a winged seed, a pencil?
- b Why do skydivers often spread their legs and arms wide when they leave an aeroplane?
- c Which factor above (shape, surface smoothness or composition) is changed when a parachute opens?
- 5 NASA engineers not only have to ensure that a spacecraft can take off in Earth’s atmosphere, they also have to make sure it can land safely back on Earth. Spacecraft have to be slowed down before they reach their destination.
- a The diagram shows a space shuttle coming in to land. Study the diagram and complete the table.

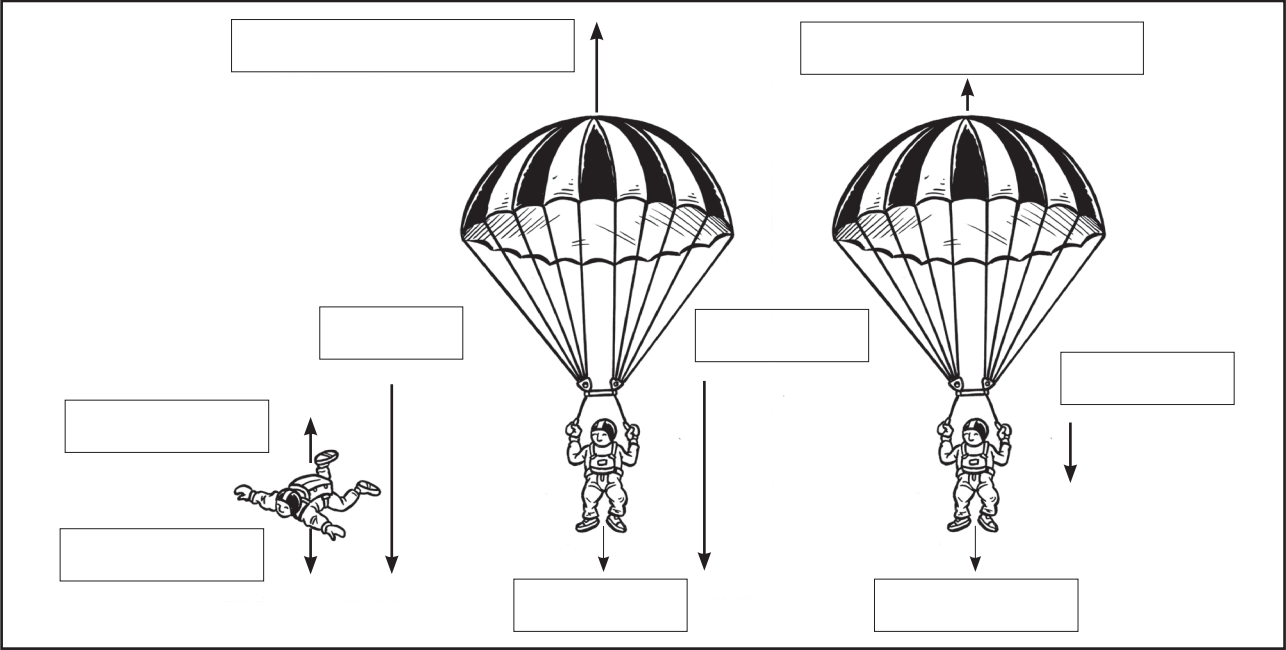


	STREAMLINED? (YES OR NO)	AIR RESISTANCE/DRAG (HIGH OR LOW)
The shuttle itself		
The braking parachute		

- b Would this system work if used on the Moon? Explain your answer.
- 6 A skydiver of known mass is moving with a force of 630 newtons downwards and a frictional air resistance of 450 newtons upwards.
- a What is the magnitude, in newtons, of the unbalanced force causing acceleration?
- b Which one of Newton’s laws explains this resultant acceleration?
- c When will the skydiver fall at a constant speed?
- d Which one of Newton’s laws explains this?
- 7 Acceleration is measured in metres per second squared (m/s^2). Acceleration due to gravity is 10 m/s^2 , as long as there is no air resistance. This means that for an object falling with no air resistance, the speed of velocity after 1 second is 10 m/s^2 , and after 2 seconds it is 20 m/s^2 , and so on.
- a What is the velocity of this object after 5 seconds?
- b If we dropped a ping-pong ball and a football at the same time in a vacuum, which would reach the ground first? Explain your answer.
- c Which famous scientist worked this out?



8 Study the diagram below of three skydivers, one of whom has no parachute. Note the ‘key’ – the size of the arrows represents the magnitude of the forces and the speed. Insert the correct term into each of the nine boxes. The terms are: small air resistance (occurs twice); bigger air resistance; high speed (occurs twice); weight (occurs three times); low speed.



9 To research the concept of acceleration, a scientist attached an odometer (records distance) and a speedometer (records speed) to a ball and then dropped the ball from a high cliff. The instruments recorded the data in the following table. Study the data and answer the questions.

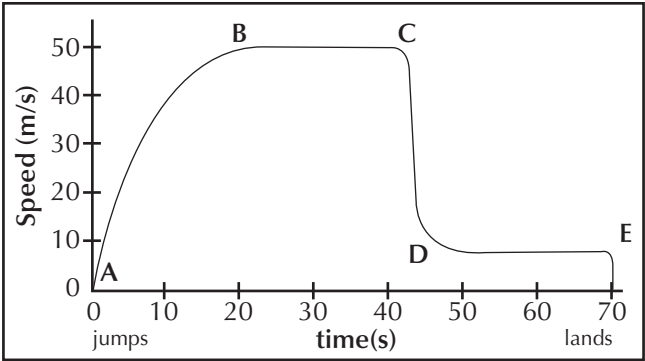
ELAPSED TIME (S)	ODOMETER (M)	SPEEDOMETER (M/S)
0	0	0
1	5	10
2	20	20
3	45	30
4	80	40
5	125	50
6	180	60

- a By how much does the ball’s speed change each second?
- b Does the distance the ball travels each second change or stay the same?
- c Is the ball accelerating as it falls? What evidence do you have?

10 Extend your understanding

A parachute is designed to have a lot of drag. As a parachutist falls down, she transfers her gravitational potential energy to kinetic energy and to heat, which warms the air. This graph illustrates speed (m/s) in relation to time (s), from when she jumps from the aeroplane to when she lands safely. What happens at the following locations on the graph?

- a Between A and B?
- b Between B and C?
- c At C?
- d Between C and D?
- e Between D and E?



★ Countdown to lift off

This simulates a multi-stage rocket launch. Provided the string is kept very taut, this system works well. A variation could be to attach a stringline to the ceiling and try a vertical launch.

★ The challenge

The force of the air rushing in one direction sends the 'rocket' in the opposite direction. Students must not only consider the best way to fold or crumple the paper, but also how to attach it to the balloon. The way the paper is attached to the balloon, and the balloon to the straw, must enable the balloon to shrink as the air escapes. For instance, the balloon should not be taped along the full length of the straw. The fishing line should be very tight,

to reduce friction. The rate of travel of the balloon will be calculated as part of this challenge: see Criteria 10 on p.6.

LINKS WITH THE EVERYDAY – space travel, popcorn, fireworks, gas turbines that generate electricity, jet engines, garden hoses, war missiles, launching satellites, safety flares at sea.

★ Curiosity trap

These two demonstrations link directly to the challenge and also to how rockets and jet engines work. You may like students to carry out these activities, but ensure appropriate safety precautions are in place for 2.

1 Blow up a balloon and let it go without tying it off. Note the direction the balloon travels. This can be repeated several times. The forward propulsion of the balloon is due to the molecules inside the balloon pushing on the wall of the balloon – not to them rushing out the rear of the balloon and pushing on the outside molecules of air. The air blows one way and the balloon goes the other way. A balloon with air escaping from it is very much like a rocket. As the air escapes from the

neck of the balloon, it produces a backwards force that pushes the balloon forwards – Newton's Third Law of Motion.

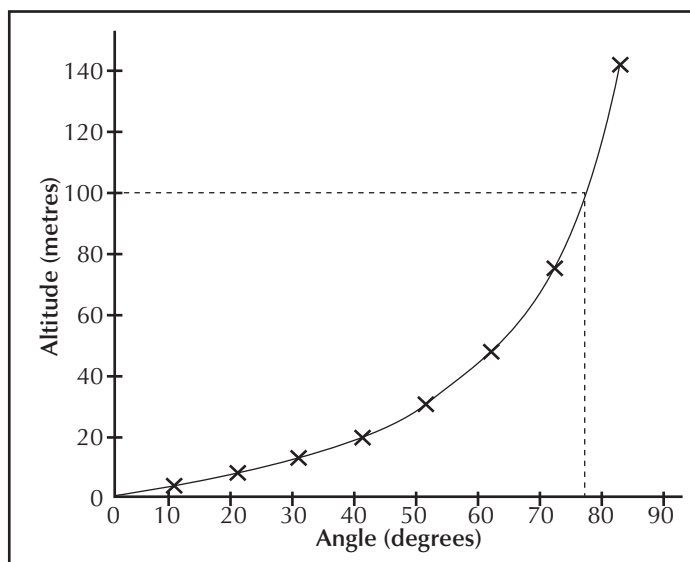
2 Take a film canister and position half an Alka-Seltzer tablet or a quarter of a teaspoon of baking powder in the upturned lid. Now **very quickly** half-fill the canister with vinegar, put the lid on, give it a shake, then set it upside down on its lid. The vinegar and baking soda react to form a gas that becomes condensed in the small space of the canister, building up air pressure. The lid blows off and the canister flies up into the air, much the same as a rocket. Wear eye protection and do not lean over the canister once it is in position.

★ Answers

- 1** **a** For every action there is an equal and opposite reaction. **b** no – there are always two objects involved
- 2** **a** Balloon (i) will not move; balloon (ii) will move to the right **b** The boat will move to the left
- 3** **a** false **b** The rocket pushes gases out the back and, in turn, the gases push the rocket forward. This process works regardless of where the rocket is **c** Rockets work better in a vacuum like outer space than in our atmosphere. This is because the exhaust gases do not have to do the extra work of pushing air out of the way
- 4** b, c, a, d
- 5** The water vapour builds up pressure, which causes an explosion. The hull of the corn bursts with quite a bit of force and the corn flies into the air
- 6** The gases push forwards on the rocket and the rocket pushes backwards on the gases, which then move backwards out of the rocket
- 7** **a** The 'action' force is applied by the squid backwards on the discharged water. The 'reaction' force of the expelled water sends the squid in the opposite direction **b** Water jetting out the end of the hose is the 'action', and you can feel the hose being pushed back in 'reaction'. With a fire hose the 'reaction' is so strong that several people are needed to hold it
- 8** **a** 40 000 km/h **b** Apollo 11 **c** Rockets need no air – they get their thrust from the reaction between their fuel and liquid oxygen
- 9** **a** The air pressure inside the bottle increases as air is pumped in. The force of the air becomes so great that the water pushes onto the bung with such force that it pops out, with the water following in a downwards direction.

As the water is pushed out, the bottle is propelled upwards in the opposite direction **b** The total weight of the rocket must be balanced by the force created by the air pressure that can be achieved inside the bottle **c** approximately 75 degrees (refer to graph below)

- 10** If a rocket is launched from the equator towards the east it already has the advantage of (about 6% of the velocity: 1000 mph) that it needs to achieve orbit and because it will be over the Atlantic Ocean if something does go wrong there will be less damage on the ground. If it were fired to the west it would not have these advantages and would also need to carry more fuel



★ Give us a clue – countdown to lift off

What you need

3 sausage-shaped balloons, 3 round balloons, 10 m of string, 1 disposable cup, paperclips, sticky tape, scissors

What to do

1 **Design** and construct a multi-stage balloon rocket by following these instructions:

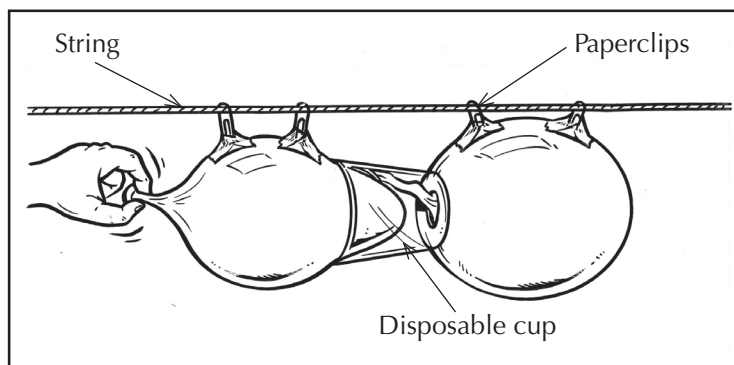
- Stretch and tie the stringline tightly between two objects in the room, for example two chairs.
 - Cut the bottom of the cup. Inflate a balloon with its end inside the cup. Inflate a second balloon with its top inside the cup so that the air pressure of this balloon seals the first balloon. Squeeze the neck of the second balloon until launch time.
 - Use sticky tape and paperclips to **attach** both balloons to the string.
- 2 To **test** your design, **release** the second balloon. Observe the flight.
- 3 **Modify** the design of your rocket system to increase the distance it will travel.

Now answer these questions:

- Are some balloon shapes better than others?
- How did you modify the rocket so that it travelled further?
- Does the weight of the balloon rocket affect its speed?
- Does the position of the sticky tape affect the operation of the rocket?
- Think about how you would explain to a friend how and why your design works.

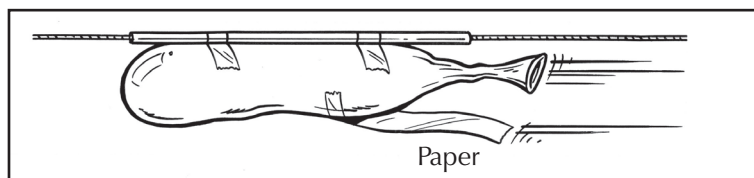
What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!



★ The challenge

Design and construct a balloon rocket that will carry an A4 sheet of paper 10 m along a line as fast as possible.



Materials

3 sausage-shaped balloons (all of equal size), 3 round balloons, 3 sheets of A4 paper, 12 m of fishing line, 3 straws, sticky tape, scissors, stopwatch, tape measure

Instructions

- 1 You have 10 minutes to work out the best way to fold a sheet of paper and attach it to the balloon and then to the straw.
- 2 You must not change your design in any way once the 10 minutes are up. **No testing is allowed.**
- 3 The fishing line may be held by two people, or the ends can be tied onto chairs or posts. This is up to each team to decide.

- 4 One team member will thread the straw onto the fishing line so that it slides easily.
- 5 At the judge's signal, release the air from the balloon. The time and distance the balloon rocket travels are recorded by the judge, allowing its rate of motion to be calculated.

Clues!

Friction is a force that acts in the opposite direction of something that is moving.

Air resistance is the friction force on something moving through air (drag); it increases as speed increases.

Speed is the rate at which an object is moving and combines information about how far and for how long an object travels.

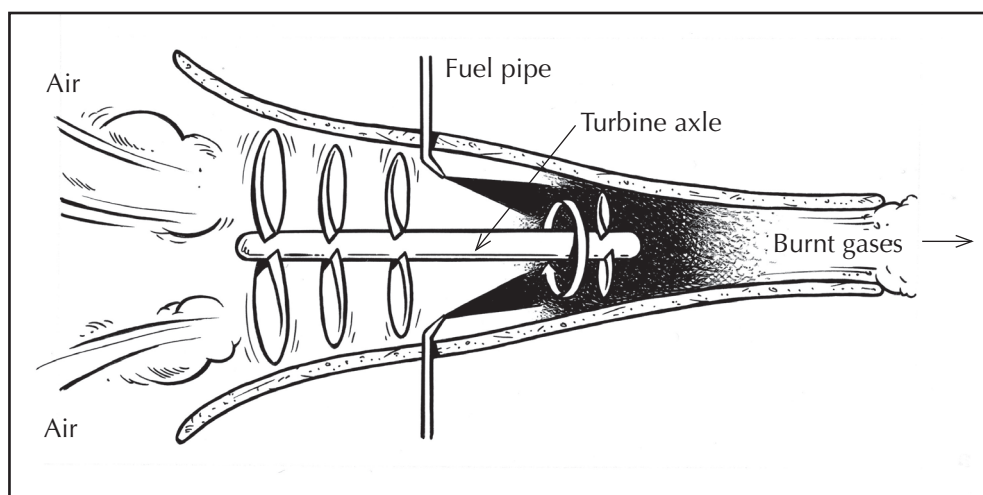
★ So what's going on?

Newton's Third Law of Motion

For every action (in this case, air escaping from the balloon) there is an equal and opposite reaction (the balloon is propelled forward). However, these two forces do not operate on the same object – the action force affects one object and the reaction force affects another. In some cases one of the 'objects' may be a gas such as air, or a liquid such as water.

Jet engine

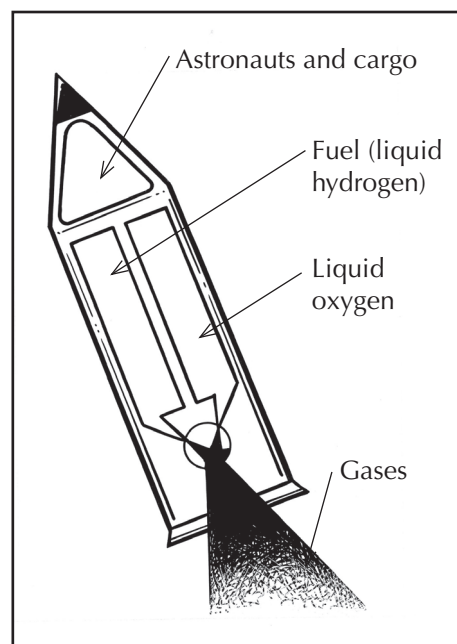
Jet engines are used on aeroplanes, and there are four stages to their operation. First a jet engine draws air into a chamber. The air is then squeezed by compressor fans as fuel is squirted in. The fuel burns continuously and the air expands. The expanded air and waste products are forced out the back at high speed and push the engine forward. The action force is exerted by the engine, backwards on the expelled air. The reaction force is exerted by the expelled air forwards on the engine, forcing the engine, and therefore the entire aircraft, in the opposite direction.



Rockets

Some of the earliest rockets were made in ancient China. Fireworks were produced using explosives that either burned quickly to make a loud bang or more slowly to propel a colourful rocket into the sky.

A rocket engine uses the same idea as a jet engine but carries its own liquid oxygen with it, so that it can travel outside the Earth's atmosphere. The rocket engine heats air or gas and pushes it out the back; in turn, the gases push the rocket forward.



Very large rockets are made in a number of stages, with each stage powered by many rocket engines. A two-stage system is often used to overcome the force of Earth's gravity. A spacecraft must accelerate many times more than the amount of acceleration that is due to gravity in order to reach orbital velocity (stage one). To reach escape velocity (stage two) the craft must fire a second time to make it go even faster. The speed needed to break away from Earth's gravity is about 11 kilometres per second or 40 000 km/h. The first manned landing on the moon was in 1969 by Apollo 11.

Explosions

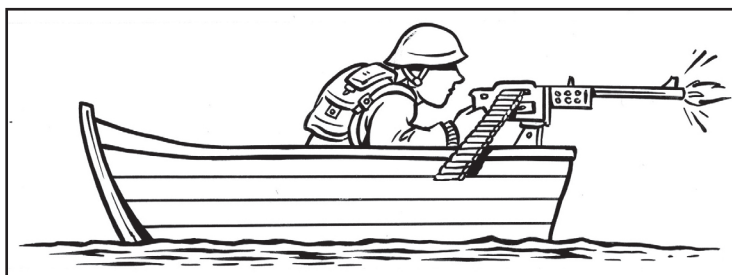
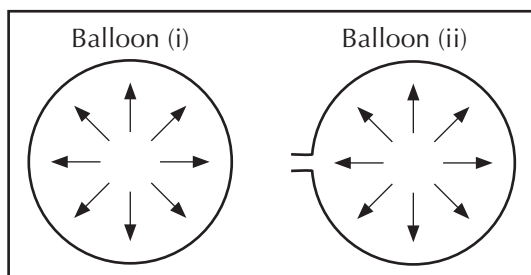
An explosion is the opposite of an implosion. In an explosion objects move apart; in an implosion they come together. The operation of a rocket is based on a controlled explosion. The rocket moves one way and the hot gases the other way. The rocket's gain in momentum is equal and opposite to the momentum of the hot gases that are ejected. Big rockets work in much the same way as fireworks and gunpowder. Gases expand rapidly in a small space and accelerate the rocket or bullet to great speed. The gases push forward on the rocket and the rocket pushes backward on the gases, which forces them backwards out of the rocket. When a rifle is fired, the rifle acts on the bullet (which goes forward) and the bullet acts on the rifle (which recoils). The operation of petrol engines also depends on the compression and explosion of petrol vapour and expanding gases.

The last word

Getting along the fishing line is only half the problem. What system would get the rocket back again?

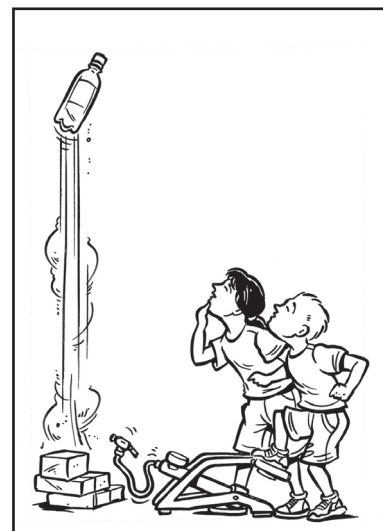
★ Figure it out!

- 1 a State Newton's Third Law of Motion.
b Do the 'action' and the 'reaction' forces act on the same object?
- 2 a Study diagram A. How will each balloon move?
b In which direction will the boat in diagram B move when the gun is fired?



- 3 Rockets don't work if you use them out in space.
a Is this statement true or false?
b Explain your answer to (a).
c Would rockets work in a vacuum?
- 4 The following information explains the four stages involved in the design and operation of a jet engine. Arrange the information in the correct sequence.
a Fuel burns continuously in the air and the resulting hot gases expand rapidly.
b Air is drawn into the engine by fans.
c Air is compressed and fuel is injected.
d Expanded gases leave the back of the engine, exerting a reaction force on the engine and pushing the aircraft forwards.

- 5 Popcorn and rockets have a lot in common. When you heat kernels of corn, you heat the white fibres of starch inside the tough yellow coating. The small amount of water turns to water vapour.
Apply your knowledge of jet propulsion to explain why popcorn pops.
- 6 Why do the gases and waste products generated by the combustion of rocket fuel move backwards out of the rocket?
- 7 Explain the following in terms of Newton's Third Law of Motion.
 - a A squid is a marine animal with a body length ranging from about 3 cm up to 6 m. It propels itself along by taking in water and expelling it in sudden spurts.
 - b What happens when someone turns a garden hose on full on while you are holding it? Why must several people hold a fire hose?
- 8
 - a How fast must a spacecraft travel to escape the Earth's gravity?
 - b What was the name of the first manned landing on the Moon?
 - c What makes rockets the best engines for spaceflight?
- 9 Rockets work by Newton's Third Law. Some children were having fun firing a water rocket straight up into the air. Study the diagram and answer the questions.
 - a Explain in terms of Newton's Third Law how this rocket works.
 - b In what way do you think the amount of water in the bottle will make a difference to the height achieved?
 - c Study the data in the table below and draw a graph of altitude (metres) vs angle of launch (degrees). What angle does the rocket need to be launched at to achieve a height of 100 m?



Data table

ANGLE (DEGREES)	ALTITUDE (METRES)
0	0
10	4
20	9
30	14
40	21
50	30
60	43
70	69
80	142

10 Extend your understanding

Consider the information provided in the box and explain why rockets launched by the United States are fired towards the east and from Florida.

Florida is located slightly north of the equator and on the western edge of the Atlantic Ocean. Very high velocities are required for space shuttles and other rockets to reach orbital velocity – typically around 27 000 kph. At the equator, the surface of the Earth is travelling at about 1600 kph towards the east.

★ Shattered alibis

Forensic science offers opportunities to use a wide range of scientific skills, such as the scientific method and analytical techniques. The word 'chromatography' means colour writing. Chemists use this process to test

a liquid mixture, such as a drug or a dye, to find out what substances are in the mixture. A Russian botanist invented chromatography to help him identify the colour pigments in plants.

★ The challenge

Students will design and perform investigations to solve a problem involving chemical separation techniques, collecting and collating data, and classifying, analysing and applying information to make deductions.

The idea is to correctly match the robber's pen with one of the suspects' pens. Students must analyse the pigments in the pen ink using the scientific method to receive the full score. An approach that is merely methodical will still receive a point. Each team must discuss their results together to come to a final decision. Five black, water-soluble pens need to be purchased and trialled before use in class. Choose

one of them as the robber's pen and include it in the bunch of five pens. Use the designated robber's pen to prepare a chromatogram in advance as a reference for the teams. The following pens contain distinctive colours: Stabilo Fine Point, OHPen 852/46; Staedtler Fine Lumocolour Non-permanent 316/9; Uni-pin fine line 03-200; Artline 803 OHP marker, Pilot Fineliner. The Pilot Fineliner has a very distinct pigment combination.

LINKS WITH THE EVERYDAY – solving crime, testing food additives, dyes in foods, plant pigments, clothing dyes

★ Curiosity trap

Take several packets of 'single coloured' jelly beans or M&Ms and mix them all up in a glass jar. Shake the jar and ask the students what sort of a mixture you are holding (a mixture of different-coloured sweets). Ask students how, and using which property, they

would separate out the different-coloured sweets into separate piles (by hand using the property of colour). Some coloured things, such as coloured molecules, are too small to separate manually – so we use a technique called chromatography.

★ Answers

1 a false b true c true

2 a The rate of migration of molecules through absorbent paper due to their differences in solubility and size b yellow c Inks and dyes consist of molecules of colouring substances that are dissolved in a liquid base. When you write, the liquid dries and leaves just the colouring molecules behind. As water creeps through the test strip, it contacts the dried colour. These molecules are loosened (dissolved) and carried through the absorbent material

3 no

4 blood, fingerprints, fibres on the glass, scratches on the metal frame

5 the maid's pen

6 a two b red and yellow c On this evidence alone, the suspect was not guilty

7 a four layers b no c no – because the colours are in a different order

8 D and C

9 suspect 3

10 Answers will depend on the situation

★ Give us a clue – shattered alibis

What you need

1 large piece of filter paper and several smaller pieces, scissors, 2 pencils, stapler, 1 cm of water in a flat dish large enough for the filter paper, access to inks of different colours (green, blue, red and yellow), a few paperclips, several M&Ms (different colours)

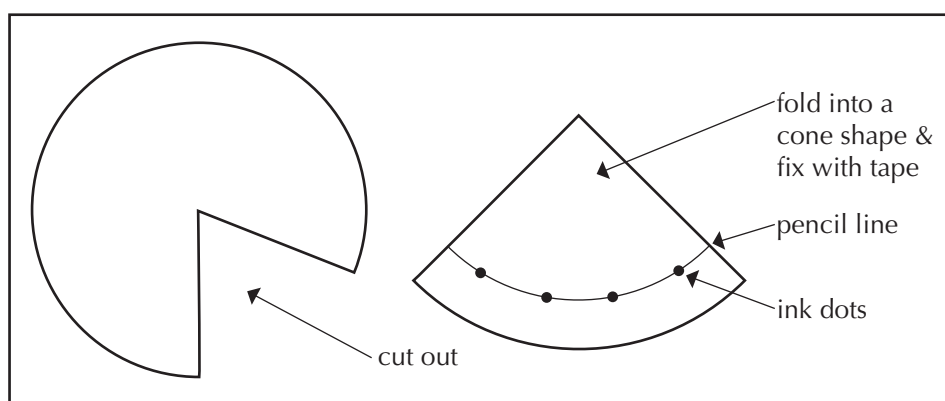
What to do

- 1 **Ask** yourself how these inks are different. Do you already know which pigments are mixed together to make different colours? What other information do you already know about inks?

Make a guess about which colours will separate out from each of the four coloured inks.

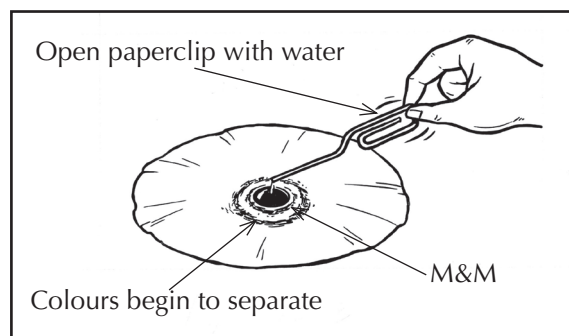
- 2 **Draw** a pencil line 2 cm from the edge of the large filter paper. Cut out a small triangle and staple the edges together in a cone shape.

- 3 **Bend open** a paperclip and use the end to put one very small dot of each ink along the pencil line, spacing the dots out evenly. Carefully place the filter paper in the dish of water, making sure the water doesn't reach the pencil line.



Observe the colours separating over the next 15 minutes as you set up the M&Ms.

- 4 **To get the colour from the M&Ms** balance a row of small filter papers along two pencils. Put one M&M in the middle of each piece of filter paper. Using an opened paperclip, slowly add 2–3 drops of water to each M&M. Turn the M&M over and add a little more water until the circle of water is about 5 cm in diameter. After a few minutes, rings of colour will form around the M&Ms.



Now answer these questions:

- Which inks are made from only one dye?
- Which pigments were used to make blue ink?
- Was your guess about the colours that would separate out from the inks correct?
- What coloured food dyes were used to make the coatings of the M&Ms?

What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

When substances are mixed together physically, as the pigments in ink are, their atoms and molecules retain their own properties and can be separated out using chromatography.

★ The challenge

Correctly identify the criminal by analysing the evidence collected from the crime scene as well as the items and information gathered by the police.

Materials

Pens belonging to the 5 suspects, water, blotting paper, a pencil, a few paperclips and paddle-pop sticks, several beakers, paper towels, a chromatograph made from the ink on the bank robber's note (available for reference)

The media report

A bank robber entered a bank and handed a note to the teller which said, 'Give me all your money.' The robber escaped with the money but left the note behind. The police later arrest five suspects: A, B, C, D and E. Each suspect's pen is impounded.



Instructions

Your task is to complete a scientific investigation of the evidence supplied and determine which of the five suspects committed the crime.

Clues!

Inks and dyes consist of molecules of colouring substances dissolved in a liquid base. When you write, the liquid dries and leaves just the colouring molecules behind. When the water or solvent creeps up a test strip, it contacts the dried colour. The ink molecules are loosened (dissolved) and carried through the absorbent material at different rates.

The rate of migration depends on solubility and size. Small, light, soluble molecules

travel further than larger, less soluble ones. Eventually the substances remain as distinct spots or bands on the paper. This colour separation is called 'chromatography'.

Chromatography can be used to separate several known pure substances from one unknown substance. The positions of the different components of the unknown substance are compared to those of the known substance.

★ So what's going on?

Murder, mutilation and grisly finds

A crime is just a puzzle to be solved. Forensic science uses scientific methods to solve crimes. 'Forensic' means carried out in a legal context. Forensic scientists use an enormous variety of analytical techniques to enable them to collect and analyse clues from a crime scene. They specialise in such areas as chemistry, biology, physics, accounting, engineering, anthropology, ballistics and even computer crime. A lot can be learned from the marks criminals leave behind.

The scientific section of a police force uses science and scientific methods and techniques to investigate crimes; the aim is to provide the law courts with impartial evidence. The Australian Federal Police's forensic laboratory in Canberra is one of our largest.

Scientific method

Science is about solving problems by understanding how everything in the universe works, including criminals. It's almost like a game with special rules called 'the scientific method'. Whenever you play this 'game', remember that the rules are flexible; they provide a general plan only, and usually leave out important bits like the hunches, the intuition and the ideas that didn't work out along the way. You can follow the scientific method by:

- 1 asking a question.
- 2 gathering information about the question.
- 3 forming a hypothesis (making a guess about the answer).
- 4 testing the hypothesis.
- 5 telling others what you found out.

Scientists tend to ask a lot of questions. They also observe, communicate, test things, look for patterns, compare things, keep records, think about things and find out new information. They work in organised ways and usually have a lot of fun.

Leaving clues

Scientists believe it impossible for someone to commit a crime without leaving something behind or taking something with them.

The discipline of forensic science is based around three fundamentals:

- 1 **Locard's principle:** When any two objects come into contact, there is always a transfer of material between the objects.
- 2 **The need to form a connection:** Are there enough powerful clues to connect the suspect with the crime scene?
- 3 **The theory of probabilities:** the basis by which evidence is weighed up. It allows the jury or judge to form an opinion as to whether there is enough evidence to prove guilt.

Some examples of clues are footprints and tyre marks, soil in shoe treads and tyres, blood, finger and lip prints, paint flakes, hair and skin, DNA, bullets and guns, and glass fragments.

Forgery

Forgery is a non-violent crime where documents are falsified in order to deceive people and gain money. Forgers sign other peoples names, make counterfeit bank notes, and alter or forge cheques by adding extra words and numbers.

Different pens contain different inks. Inks are made of a mixture of dyes, and when separated out they can help identify which pen was used in a crime. Criminals may write ransom notes, instructions to victims, burglary addresses, the names of those murdered, and so on.

Recently there has been an increase in computer crime, where 'hackers' find out confidential information such as bank account details. Computer crime is extremely costly to the community.

The last word

Humans and koalas have almost identical fingerprint patterns!

★ Figure it out!

1 Are the following statements true or false?

a Chromatography is particularly useful if there are very large amounts of substances in a mixture.

b Chromatography was first used to separate plant pigments (colours).

c Chromatography can be used to identify the components of an unknown mixture.

2 To separate substances, we must look for differences in the physical properties of the individual components. The diagram shows the results of a chromatography test done on black ink with water.

a What differences are we employing here?

b Which pigment is the most soluble in water?

c Explain the chemistry of this test.

3 Some people are allergic to the dyes in foods and should not eat them. Certain iceblocks have added colours, such as 'colour 133'.

Study the results of the test. Should a person allergic to 'colour 133' eat this iceblock?

4 A thief entered a room by breaking a window and then forcing the metal window frame with a screwdriver. As he climbed in, he cut his hand. What are three pieces of evidence the police might look for?

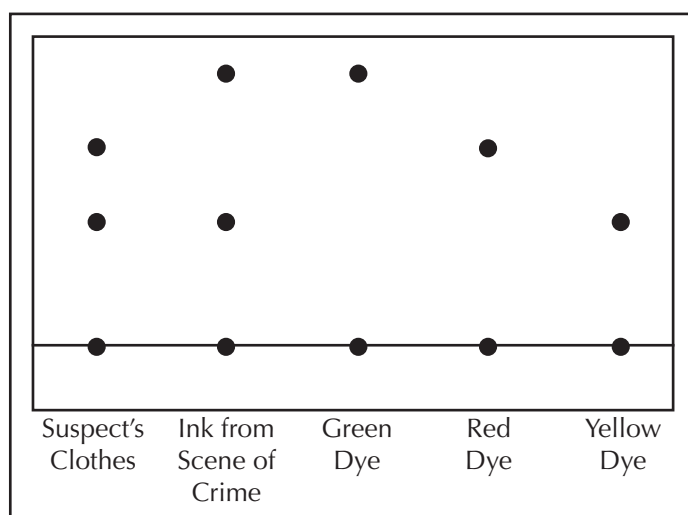
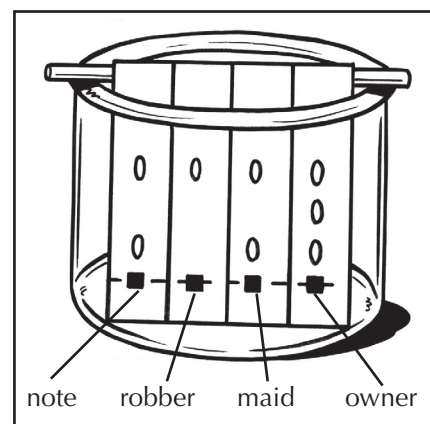
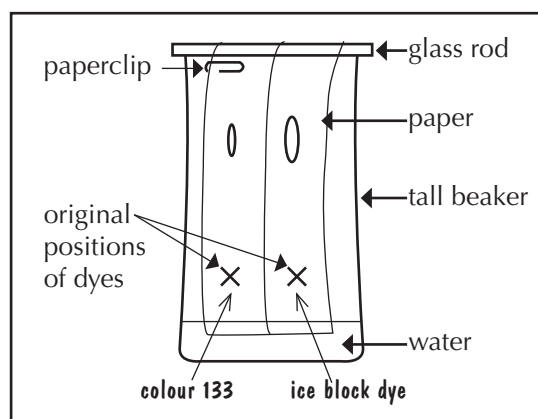
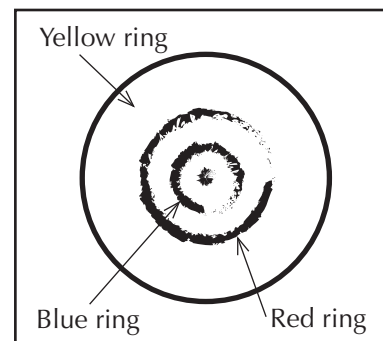
5 Police found a piece of paper with the address of a crime scene written on it. The forensics department carried out a test to find out if the address was written by the robber's pen, the owner's pen or the maid's pen. Study the results and decide whose pen was used to write the note.

6 A thief broke into an office at night and knocked over a bottle of ink, spilling some on her clothing. A suspect was later arrested and forensic scientists compared the ink on her clothing with that remaining in the bottle and with three pure ink samples. The diagram shows the results.

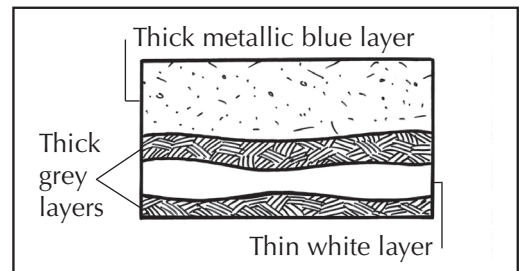
a How many inks were in the suspect's sample?

b What colours were in the suspect's sample?

c Based on this evidence, was the suspect guilty or not guilty?



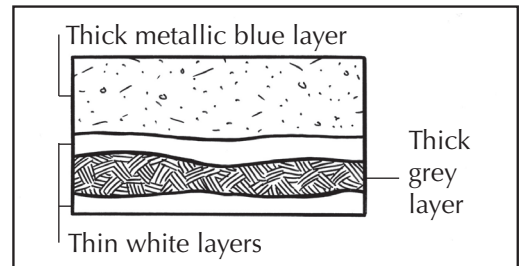
- 7 Paint can be collected from the crime scenes of hit-and-run car accidents and forced entries. The paint can be compared microscopically with other samples, especially if all the paint layers are present. The more coats of paint that match, the higher the probability they came from the same source. The diagram shows two paint chips, one collected from a hit and run site and one from the suspect's vehicle.



a How many different layers of paint can you see in both samples?

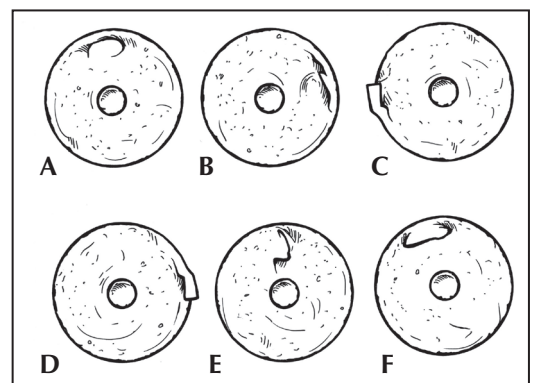
b Are the layers all the same thickness?

c Is this vehicle likely to be the one involved in the crime? Explain your answer.



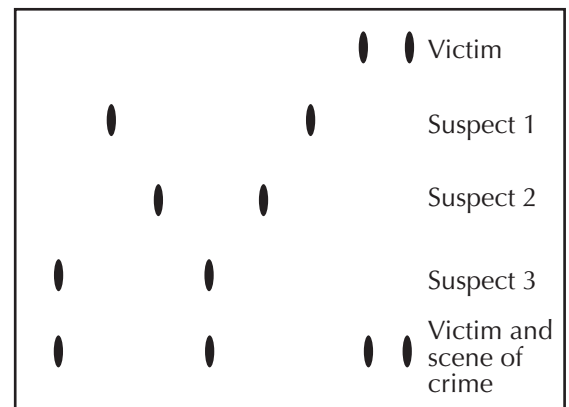
- 8 A ballistics expert would note the position of the victim, the position of the firearm, the position of any spent cartridges and where any bullets are lodged. Bullets and gun barrels are compared in the laboratory. Each gun barrel has unique grooves that make marks on the bullets fired. All bullets fired from the same gun will have identical marks.

Study these six drawings of .22 inch rim cartridges. Which could have been fired by the same gun?



- 9 DNA is the genetic map or code that determines the characteristics of each human being. Apart from identical twins, no two people have the same DNA patterns. DNA can be collected from crime scenes in the form of skin cells, hair, and body fluids such as blood, saliva and semen. A forensic scientist would complete a DNA profile of the criminal from samples collected at the crime scene and compare it with a DNA profile of the suspect. DNA code is separated out using an electrical current through a pad of gel agar.

The diagram compares the genetic profiles of a murder victim, the crime scene and three suspects. Based on this evidence only, which suspect is the murderer?



10 Extend your understanding

The three fundamentals of forensic science are that (1) any contact leads to a transfer of materials and (2) provides clues, but (3) we need to determine whether there are enough clues to prove guilt.

Scientists believe that it is impossible for someone to commit a crime without leaving something behind or taking something with them. Observation is a very important scientific skill and invaluable in forensic science for collecting all possible clues from a crime scene. Sometimes the most obvious clues are overlooked. Without looking up from this page, answer the following questions.

a Briefly describe the room you are in.

b Without looking, describe the hair of the person next to you.

c Without looking, list any significant characteristics of the person on the other side of you.

d Without looking, describe your teacher, including what they are wearing, and list their jewellery.

e What kind of animal was wearing a coat and holding a magnifying glass in the sketch on the page explaining the Challenge? Was it wearing a hat?

★ Sizing it up

Students will use prediction, measuring and recording to rank a series of regularly and irregularly shaped objects. The findings for the class could be combined and displayed to allow students to complete a brief report on the methods used in calculating the volumes.

Examine the idea that two objects cannot be in the same place at the same time. Assuming that there will be some error in the measurements, have students discuss the possible sources of error. Further information on density can be found in 'Robust rafts', in Book 1 of this series.

★ The challenge

The packages must be tightly wrapped and of different sizes; students should not be able to feel too much through the packages. They know that a number of the packages are made out of the same substance, even though the packages have different shapes. Each group must have 8 to 10 mystery packages, labelled A, B, C, etc. Packages of the same substance must vary in mass. Suggested substances are marbles, limestone chips, modelling clay, sand, popcorn or 5-cent coins. Make sure that the volume of every package can be found using the measuring cylinder you provide.

To cut down on measurement errors, the smallest package should displace at least 15 mL of water.

The objective is to identify the substance by focusing on the mass and volume of each package and realising that there is a constant mass/volume relationship for any substance made out of the same material. Neither the mass nor the volume are important on their own, it is the mass/volume relationship that identifies the substance.

LINKS WITH THE EVERYDAY – builder's level, salad dressing, fresh eggs sink but old eggs float.

★ Curiosity trap

Carry out the following three activities in order and encourage students to assist with the calculations.

- 1 Calculate the volume of a 500 g pack of butter (volume = length × breadth × height).
- 2 Calculate the volume of the non-uniform shape of six fresh eggs. The volume can be found by putting all six eggs into a large beaker and recording the volume of the displaced water by subtracting the initial water level from the final

water level in the beaker. Calculate the average volume of an egg (average volume = total water displaced divided by 6)

- 3 Calculate the density of an egg. Point out that to calculate the density of a substance, we must first know the object's mass or weight in grams as well as its volume in mL. Then we divide the mass by the volume. The average mass of an egg (in grams) can be found by weighing all six eggs on a set of scales and dividing the result by six.

★ Answers

- 1 **a** density **b** kilograms per cubic metre or grams per millilitre per cubic centimetre **c** the amount of matter in a substance **d** Density = mass divided by volume
- 2 **a** Density = mass divided by volume = 0.8 g/cm³
b Mass (g) = density (g/cm³) × volume (cm³) = 50 g
c Volume = mass divided by density = 25 mL
- 3 mass, density, mass/volume, constant
- 4 **a** Two different kinds of matter cannot be in the same place at the same time **b** The flask is not empty, it is full of air
- 5 Pour water into the jar of marbles, which replaces the air with water and forces the air out of the jar. Since we are interested in the amount of air there was before the air was forced out, we only need to measure the volume of water needed to fill the jar
- 6 The coins are weighed and the total weight compared to the established weight of a certain value of coins

- 7 **a** The bird took a large breath and blew all the air it could into the bottle. The volume of air in the bottle can be measured and is equal to the bird's lung capacity **b** water
- 8 **a** Because there is equal weight at each end of the ruler **b** The finger displaces some water, so the water level in the glass rises as the finger takes the place of some of the water. The water the finger replaces is still in the glass, so the overall weight of the glass has increased and the ruler tips
- 9 The substances that would sink are the iron filings, ice cube and salt water; those that would rise are the cooking oil and alcohol. Water will remain suspended in the hole as its density is the same
- 10 **a** The gas would be less dense than air **b** If the mass of 1.0 L of the experimental gas is heavier than 1.29 g it will fall (carbon dioxide, butane and oxygen). If it is less than 1.29 g it will float upwards (methane and helium)

★ Give us a clue – sizing it up

What you need

6 objects (a ream of paper, a birthday candle, a spoonful of sugar, 5 cm piece of rope, 5 marbles, a film canister of olive oil), a ruler, 100 mL graduated cylinder, a large beaker and a smaller beaker that will fit inside it, water, a bottle of rubbing alcohol

What to do

- 1 **Guess** the volume of each of the objects provided and record these predictions in the data table.
- 2 **Read the ‘Tips’** that follow and then, using the materials provided, measure the actual volume of each object and record your results in the data table.

Tips

- Sometimes indirect methods are required to find the volume of a single object.
- The **birthday candle** will sink in water but float in alcohol.
 - The **sugar** will dissolve in water but not in alcohol.
 - The **volume** of a single marble or a single piece of paper can be found by dividing the total volume by the number of single marbles or sheets of paper in the ream.
 - The **density** of alcohol is much less than water.

Data table

OBJECT	PREDICTED VOLUME (cm³)	ACTUAL VOLUME (cm³)
Ream of paper		
Birthday candle		
Spoonful of sugar		
5 cm piece of rope		
5 marbles		
Film canister of olive oil		

Now answer these questions:

- Compare your predictions with the actual volumes measured.
- Rank the objects from smallest to largest volume.
- Calculate the volume of one piece of paper.
- Calculate the volume of one marble.
- Why did you need to use the alcohol instead of the water for some measurements?

What happens next?

After you have completed this experiment and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

Two different kinds of matter cannot be in the same place at the same time.

The **volume** of liquid displaced equals the volume of the object below the liquid’s surface.

Displacement is when one substance replaces another by pushing the first one somewhere else.

★

The challenge

Without opening, removing or tearing the aluminium foil, work out which of the mystery packages are made of the same substance.

Materials

8 numbered mystery packages wrapped tightly in aluminium foil, a balance, a large graduated cylinder, a supply of water

Instructions

- 1 Using the equipment provided, work out which of the mystery packages contain the same substances. If you spend time feeling the packages for clues or burst the packages in any way, your team will be disqualified.
- 2 Record your results in the table provided below. An extra column has been provided. Use it to record the answer to any other calculations that may be necessary. Label the extra column, including the unit of measurement, then fully analyse all the data to reach your conclusion.
- 3 Write your conclusion in the space provided under the table.

Data table

MYSTERY PACKAGE	MASS (G)	VOLUME (ML)	
A			
B			
C			
D			
E			
F			
G			
H			

Conclusion: _____

Clues!

Mass is the amount of matter in a substance.

Volume is a measure of the amount of space occupied by the matter in a substance.

Density measures the packing of particles in a substance. The density of a substance equals its mass divided by its volume – its mass/ volume ratio.

*The **mass/volume** ratio can be used to identify substances.*

☆ So what's going on?

Measuring

Measurement is comparison. Scientists use standard units to compare objects. For instance, the standard unit for length is a metre, for volume it is a litre, and for mass it is a gram or a kilogram.

Two types of measurement error can occur in scientific experiments. Measuring instruments are only accurate to a certain degree, and different people may read the scale on a particular instrument slightly differently. Scientists overcome these kinds of errors by repeating an experiment several times and by using averaging techniques.

Measuring mass

The **mass** of an object is how much matter it contains. **Matter** is collections of different kinds of atoms which take up space and can be weighed. Mass is measured in kilograms (kg). To find the mass of an object, simple balancing scales are used to compare the unknown mass with a known mass.

Measuring volume

The volume of an object is the amount of space it occupies. Volume is measured in cubic metres or cubic centimetres. Note that one cubic centimetre equals 1 millilitre.

The volume of regular-shaped solids can be found using a ruler and a mathematical formula. For example, the volume of a rectangular block is found by using this formula: length $\times$ breadth $\times$ height. The volume of a solid with an irregular shape is measured by displacement. If an object is dropped into liquid, it displaces or 'pushes aside' some of the liquid, making the liquid level rise. This volume of liquid displaced is equal to the volume of the object and can be measured in millilitres or in cubic centimetres.

The volume of an object cannot always be measured using such direct methods. **Indirect methods** must sometimes be used to find the volume of a single object. There is a constant relationship between the mass and the volume of any substance made of the same material.

Measuring density

Objects of the same size and shape can vary greatly in mass. For example, one cubic centimetre of cork is much lighter than one cubic centimetre of lead. The molecules of lead are heavier and more closely packed together than those of cork. This makes lead a denser material than cork. A substance with a low mass and a large volume, such as a gas, has a low density. Every substance has its own density or mass/volume ratio, which helps us in identifying substances. To find the density of a solid, a liquid or a gas, its mass and volume must be measured and used in the following formula:

$$\text{Density (d)} = \frac{\text{Mass (m)}}{\text{Volume (v)}}$$

Density is measured in kilograms per cubic metre, or grams per cubic centimetre. Substances with densities greater than the density of the liquid in which they are placed will sink. Substances that are less dense than the liquid in which they are placed will float.

The last word

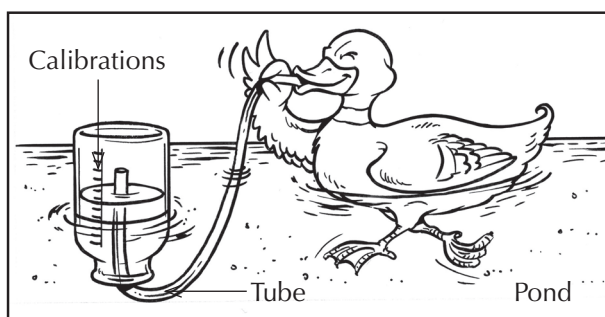
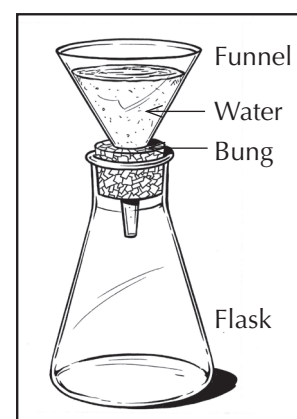
Why do ice cubes (solid) float on top of water (liquid)?

★ Figure it out!

- 1 a What is the name given to the ratio of mass to volume?
b What standard unit is used to compare this relationship between substances?
c Define the term 'mass'.
d Write the mathematical formula for the relationship in (a).
- 2 a Calculate the density of a rock with a mass of 200 g and a volume of 250 cubic centimetres.
b The density of water is 1 g/mL/cm³. Calculate the mass of 50 cubic centimetres of water.
c What is the volume, in mL, of 20 g of a liquid with a density of 0.8 g/cm³?
- 3 Read the information below and fill in the missing words.

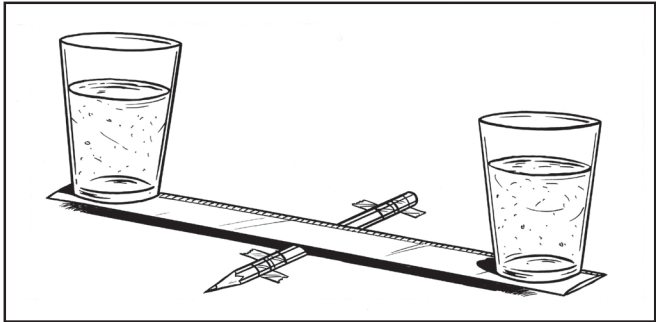
For the same volume, different substances will have a different _____. In other words, they have different _____. It is neither mass nor volume that is important, but the _____ relationship that is characteristic of a substance's identity. There is a _____ mass/volume ratio for any substance made out of the same material.

- 4 Study the diagram, which shows a funnel full of water with its shaft through a bung that has been firmly pushed into the neck of an empty Erlenmeyer flask.
a Explain why the water refuses to run into the flask.
b Is the flask actually empty? Explain your answer.
- 5 We have a jar full of marbles. How could we measure the volume of air in the jar? (Clue: We are actually interested in the amount of space between the marbles in the jar.)
- 6 Shops collect large quantities of coins, which they regularly take to banks in order to swap them into a more convenient form, such as notes. What 'indirect' method does the bank use to count large numbers of coins of the same value?
- 7 The volume of a liquid is easy to measure directly. We can use an indirect method to measure the amount of gas we can hold in our lungs. Study the diagram of a bird standing in a pond and blowing into a tube. The end of the tube is in a bottle. The bottle was calibrated, filled with water and submerged upside down in the pond, so that no water escaped.

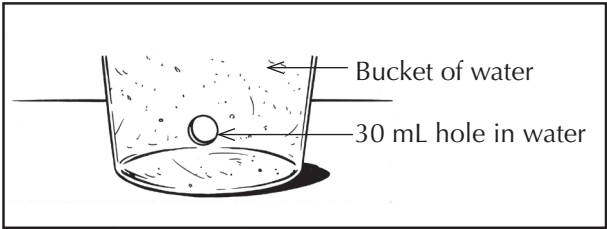


- a Explain how this bird worked out its own lung capacity.
- b What substance was pushed out of the way and replaced by the bird's exhaled breath?

- 8 The diagram below shows a ruler balanced across a pencil, and two glasses of water balanced on the ruler a bit like a see-saw. Neither end of the ruler is touching the table.
- a Why isn't the ruler touching the table?
- b What will happen if you dip your finger into the water in one of the glasses but do not touch the glass? Explain your answer.
- 9 Equal volumes of different liquids have different masses, so different liquids have different densities.



Imagine we have a bucket of fresh water with a 30 mL hole carved out of the middle of the water. Look at the data table and predict what would happen if we filled the hole with each of the substances listed. Complete the table.



Data table

SUBSTANCE (30 ML)	SUBSTANCE DENSITY (G/ML/CM³)	SINK	RISE
Iron filings	7.0		
cooking oil	0.8		
ice cube	0.9		
water	1.0		
salt water	1.2		
alcohol	0.8		

10 Extend your understanding

- a Imagine the bucket in Question 9 was full of air instead of water and we filled the hole with other gases, such as methane or oxygen. What would have to be true if the experimental gas floated upwards?
- b Imagine that you have five weightless balloons, each containing 1 litre of a different gas. You suspend them in a room full of air, which has a density of 1.29 g/L. Predict what would happen for each of the gases in the data table and tick the 'rise' or 'fall' columns.

Data table

EXPERIMENTAL GAS (1.0 L)	MASS (G) > 1.29	MASS (G) < 1.29	RISE	FALL
Carbon dioxide	*			
Butane	*			
Methane		*		
Oxygen	*			
Helium		*		

★ Change agent

Alka-Seltzer tablets contain bicarbonate of soda, citric acid and a salicylate analgesic. As long as they remain sealed in their foil packets, no reactions can occur. However, the presence of water (even humidity in the air) will induce a reaction that produces carbon dioxide gas. The rate of gas production indicates the rate of reaction. This investigation encourages students to find out about the factors that can affect the speed or rate of a chemical reaction and provides experience in the separation and control of variables.

Asking students strategic questions may draw out some of the information in italics below, but be careful not to reveal all the information as individual analysis and the resultant understanding of this concept will be tested in the challenge.

The reaction between water and Alka-Seltzer speeds up: at higher temperatures; when the concentration of Alka-Seltzer is greater; when the surface area is large (ground up tablets); and when the reaction mixture is continually agitated.

★ The challenge

If there is any doubt about who has won, the class must devise a way to measure the volume of each balloon. To do this, submerge the balloons in water and measure the volume of the displaced water. The team with the largest balloon wins. The 'accepted' scientific view of rate of reactions can be locked in

after the results of the challenge are recorded. This information is given in italics above and also in the 'So what's going on?' section.

LINKS WITH THE EVERYDAY – combustion, corrosion (rusting), digestion, oil refining, cold packs (endothermic reactions), antacid tablets

★ Curiosity trap

It would be good to complete all three activities below, but one on its own will also introduce the unit.

- 1 Make up the following mixture and put a small teaspoonful into a miniature ziploc bag for each student. Mix half a cup of icing sugar with half a bag of jelly crystals, a quarter of a teaspoon of baking powder, half a teaspoon of citric acid and half a teaspoon of tartaric acid. Have students tip this mixture onto their tongue and observe the reaction that occurs when it comes in contact with saliva (water). The mixture will fizz as carbon dioxide gas is produced.
- 2 The day before the class, take an Alka-Seltzer tablet out of its foil wrap and leave it in the refrigerator uncovered. At the time of the class, remove the tablet from the refrigerator. This tablet has been exposed to a small amount of water (humidity) and will have begun to react, but very slowly. Prepare three new tablets as follows –

leave one whole, break one into four pieces, and grind the third into a powder. Taking three small beakers containing 50 mL of tap water, drop one tablet into each and observe the speed of reaction – the time it takes (in seconds) for each tablet to completely dissolve. The amount of water makes a difference to the rate of reaction – compare the exposed tablet with those dropped into the water. The surface area of the substance also makes a difference to the rate of reaction.

- 3 To introduce the concept of biological reactions, put a 1–2 cm cube of liver into a large ziploc bag. Add about 10 mL of hydrogen peroxide and firmly close the bag. This reaction produces bubbles of oxygen gas. This chemical reaction also occurs inside our livers. Hydrogen peroxide is a waste product of the body that the liver turns into water and oxygen. This reaction also produces heat, helping us maintain our body temperature.

★ Answers

- 1 Situations 1, 3 and 5 are physical reactions; situations 2, 4 and 6 are chemical reactions
- 2 **a** temperature, concentration, surface area and mixing **b** They all enhance or restrict the chances of the reactant particles present colliding with one another and therefore increasing or decreasing reaction rates **c** paint or plastic
- 3 **a** nails 1 and 3 **b** The oil stops more oxygen becoming dissolved in the water **c** nail 2 **d** air and water **e** yes, because nail 2 rusted the most **f** oxidation
- 4 **a** enzymes **b** digestive system, respiration system
- 5 **a** the Law of Conservation of Mass **b** Although substances may change into other substances during a chemical reaction, the actual number of atoms remains

- the same **c** The original reactants are carbon and lead oxide; the new products are lead and carbon dioxide
- 6 The vinegar (acid) and the baking powder (base) reacted to produce carbon dioxide, smothering the flame
 - 7 energy; endothermic; exothermic; combustion; respiration
 - 8 **a** curve A **b** Curve A is steeper, so rate must be faster
 - 9 **a** the magnesium reaction **b** magnesium **c** iron **d** magnesium, zinc, then iron **e** zinc + hydrochloric acid → zinc chloride + hydrogen
 - 10 **a** test 1, because a catalyst was used to increase the rate **b** catalysts **c** Catalysts speed up reactions by bringing the reactants together and then moving away unchanged

★ Give us a clue – change agent

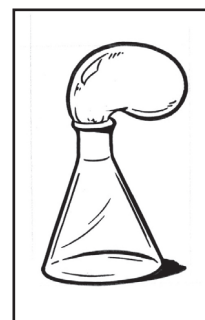
What you need

3 × 100 mL Erlenmeyer, Alka-Seltzer tablets, water (iced, hot and room temperature), plastic teaspoons, mortar and pestle, 50 mL measuring cylinder, 2 balloons

What to do

Carry out the following 4 experiments, which will provide you with useful information about chemical reactions. After each test, rinse the 3 flasks with fresh water.

- 1 Pour** 50 mL of iced water into one flask, 50 mL of tap water into another, and 50 mL of hot water into the third. Add one Alka-Seltzer tablet to each flask and observe how fast the tablets dissolve.
- 2 Take** 3 Alka-Seltzer tablets. Keep one whole, break one into quarters and grind up the third in the mortar and pestle. Pour 50 mL of tap water into each of the 3 flasks. Add the whole tablet to one flask, the broken tablet to the second flask, and the ground-up tablet to the third flask. Observe which tablet dissolves the fastest.
- 3 Pour** 50 mL of tap water into 2 flasks. To the first beaker, add 1 ground-up tablet; quickly pull a balloon over the mouth of the flask to catch the gas produced. To the other flask add 3 ground-up tablets all together and quickly attach a balloon. Observe how much gas is produced in each balloon.
- 4 Pour** 50 mL of tap water into 2 flasks. Add a whole Alka-Seltzer tablet to each one. Vigorously swirl one flask but not the other. Observe how fast each tablet dissolves.



Now answer these questions:

- At what temperature does the reaction go the fastest?
- Does the amount of substance affect the speed of the reaction?
- Does stirring increase or decrease the reaction rate?
- Which of the factors you tested were important in determining how fast each reaction occurred?

What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

A **chemical reaction** is a process in which one or more chemical elements or compounds form new compounds. All chemicals can change into something else if the conditions are right.

Physical changes **do not** make new substances.
Chemical changes **do** make new substances.

Reaction rate depends on the time taken – the faster it happens, the greater the reaction rate.

★ The challenge

Work out a set of conditions that will result in the fastest chemical reaction over a 2-minute period, using the equipment provided.

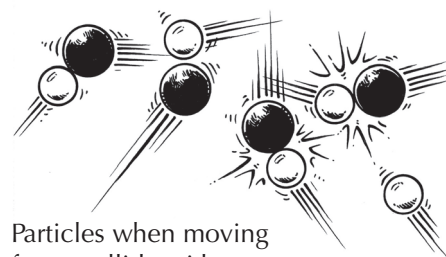
Materials

A bowl of iced water, a beaker of hot water (at least 60 degrees), 100 mL Erlenmeyer flask, one-holed stopper to fit the flask, a balloon (large enough to fit over the stopper inserted into the flask), Alka-Seltzer tablets, mortar and pestle, 50 mL graduated measuring cylinder.

Instructions

- 1 You must **put** exactly 50 mL of water into the 100 mL Erlenmeyer flask.
- 2 **Timing** begins when the Alka-Seltzer enters the flask and finishes exactly 2 minutes later.
- 3 **At the end** of the 2 minutes the balloon must be pinched off and tied. The judge must observe this procedure to ensure that no extra air is added to the balloon.

Particles in a reaction



Particles when moving faster collide with greater power and more often

Clues!

The **reaction rate** is the rate at which products are formed or reactants used up in a chemical reaction.

The greater the number of **collisions** between reactant molecules, the greater the reaction rate.

Reaction rate depends on **factors** such as substance concentration, temperature, agitation and surface area.

★ So what's going on?

Chemical interactions

All the elements on this planet, often referred to as 'chemicals', are grouped and listed in the Periodic Table, which can be found in most chemistry texts. This matter exists as pure substances such as lead or gold as well as compounds of several of these substances, such as water and carbon dioxide. All chemicals can change into something else if the conditions are right.

When something changes, we say that it is different. If the change results in a new substance, it is called a **chemical change** and can rarely be reversed. If the change does not make a new substance it is called a physical change and is easy to reverse. Chemicals can meet without reacting, but if a new substance is formed from the existing ones, a **chemical reaction** must have occurred. Some signs of a chemical reaction include the production of a gas or a solid, a colour change or a temperature change. Every chemical reaction involves energy changes – some produce heat (exothermic reactions) and others use heat from the surroundings (endothermic reactions).

Chemical reactions take place in the world around us all the time. Living things actually depend on chemical reactions. There are several different types of reactions. One example is an oxidation reaction, which occurs when a substance combines with oxygen. Some oxidation reactions are useful, such as combustion where a substance like wood or candle wax burns to release energy. On the other hand, corrosion or rusting, particularly of metals, is an oxidation reaction that can be a problem. Non-reactive

barriers, such as paint or plastic, are needed to slow down these reactions. The combination of oxygen with the carbon and hydrogen in our food to release energy is an oxidation reaction called respiration.

Rate of reaction

If particles collide more often or with more energy, the rate of the reaction will increase. The rate of collisions between reactants can be affected by things such as substrate concentration, surface area, temperature and mixing.

Chemical helpers

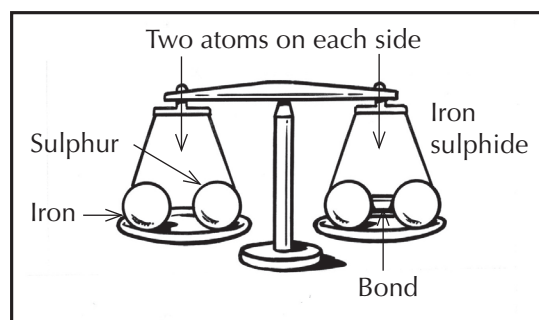
Many chemical reactions proceed very slowly, but some can be speeded up by adding another substance. A **catalyst** can speed up a reaction because it 'collects' the reacting particles, brings them together and then moves away again. The catalyst is not changed in any way; nor is it used up. It is not easy to find a catalyst for a particular chemical reaction.

A large number of catalysts, called **enzymes**, are found in living tissue. Digestive system enzymes are found in saliva and in gastric juices. These enzymes speed up the breakdown of large molecules, such as starch and protein, into simpler molecules the body can use.

Law of Conservation of Mass: Matter cannot be created or destroyed, only changed from one form to another.

The actual number of atoms in a chemical reaction remains the same. In other words, the number of atoms we start with is the same number of atoms we end up with.

Any chemical reaction can be written in an abbreviated form, called a word equation. We put the substances we start with on the left and the new substances that are formed on the right. For example: iron + sulphur → iron sulphide.



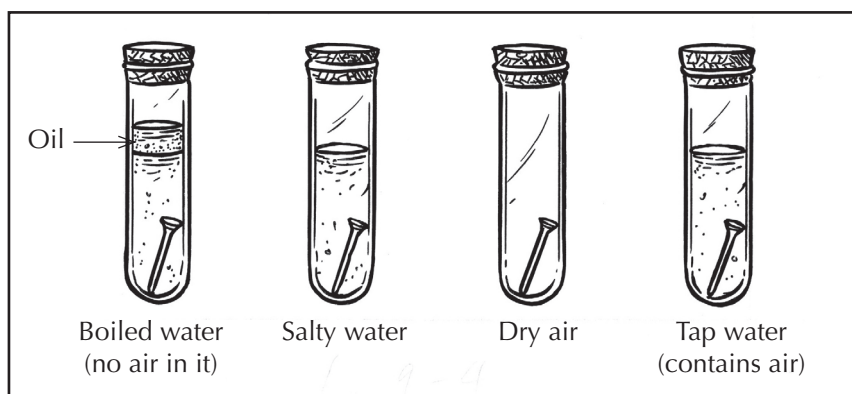
★ Figure it out!

- Physical changes do not make new substances, but chemical reactions do. Study the table below and tick which of the situations are physical reactions and which are chemical reactions.

SITUATION	PHYSICAL	CHEMICAL
1. A hot-air balloon going up		
2. A young tree growing every year		
3. The change in a thermometer as the air around it warms up		
4. A rocket taking off		
5. Ice melting into water		
6. The combustion engine of a car		

- The rate of a reaction can be affected by a number of factors.
 - List four of these factors.
 - How do these factors affect reaction rates at the molecular level?
 - Give an example of a substance that would slow down the reaction between iron and air.

- 3 An experiment was set up to investigate why things rust. Iron nails were placed in four different liquids. Study the diagram and the results table, then answer the questions.

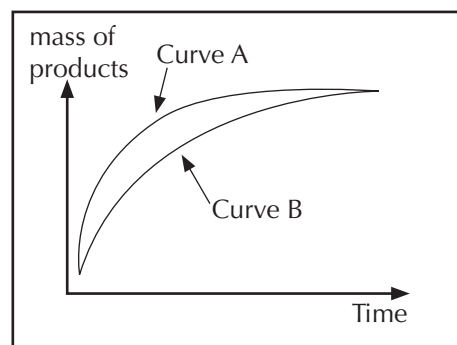


TEST TUBE NUMBER	NO RUST	RUSTY	VERY RUSTY
1	✓		
2			✓
3	✓		
4		✓	

- a Which nails did not rust?
- b Why do you think oil was put on top of the boiled water?
- c Which nail rusted the most?
- d What two substances are required for iron to rust?
- e Cars seem to rust faster near the sea. Does the data here support this idea?
- f What is the name given to this type of chemical reaction?
- 4 A large number of catalysts are found in living tissue and act by speeding up or slowing down chemical reactions in the body.
- a What are these biological catalysts called?
- b Give two examples of systems in the body where such catalysts operate.
- 5 A basic law of chemistry is that you cannot create or destroy matter in a chemical reaction.
- a What name is given to this law?
- b Explain what the law means.
- c Identify the new products and the original reactants in this word equation:
carbon + lead oxide → lead + carbon dioxide
- 6 A group of students were investigating chemical reactions. They stuck a candle in the centre of a glass dish with sides taller than the candle. They sprinkled baking powder around the base of the candle and then lit the candle, which burned brightly. After pouring some vinegar around the candle, they observed that bubbles were produced and the candle flame went out.
Explain the results of this investigation.
- 7 Complete the information below by filling in the missing words.
Chemical reactions involve _____ transfer. In most chemical reactions heat is either given out or taken in. If a chemical reaction uses heat it is called an _____ reaction. If a reaction produces heat it is called an _____ reaction. Two examples of this type of reaction are _____ and _____.

- 8 The rate of reaction can be affected by temperature. If the reactants are heated, the particles move faster and there is a greater chance they will collide with one another. The rate of the reaction is therefore faster.

Study the graph and answer the questions.



a Which curve shows the reaction that occurs when the reactants are heated?

b What evidence on the graph leads you to this conclusion?

- 9 A science class was studying the reaction between dilute hydrochloric acid and the metals zinc, iron and magnesium. They put the same amount of metal and acid in each flask and attached a balloon to the neck of each flask to catch any gas produced. Study the diagram and answer the questions.

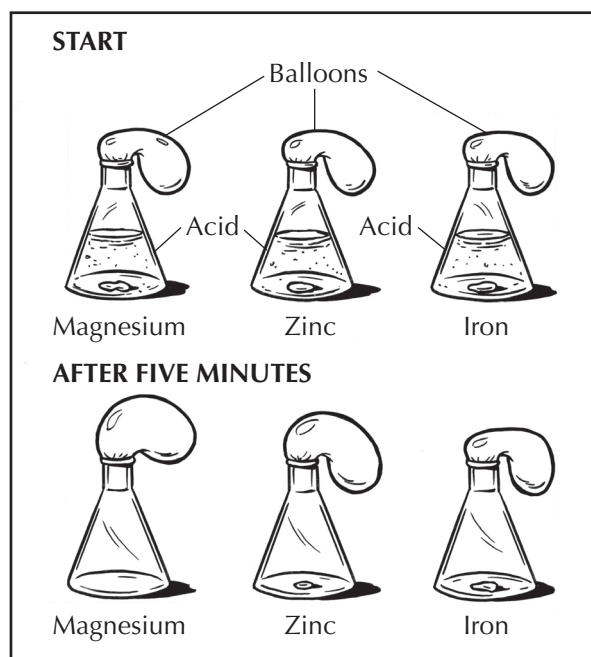
a Which reaction produced the most gas?

b Which metal reacted the fastest?

c Which metal reacted the slowest?

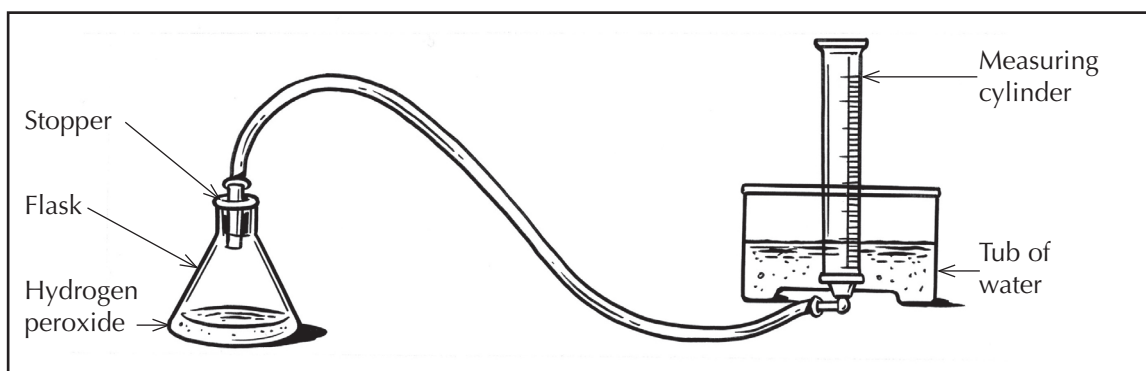
d Write down the metals in order of their reactivity.

e When zinc reacts with hydrochloric acid, two new substances are produced – hydrogen and zinc chloride. Write the word equation for this chemical reaction.



10 Extend your understanding

A student was curious about substances that affect the rate of some reactions and carried out two tests. For test 1, he set up the equipment in the diagram below. He poured 10 mL of hydrogen peroxide and 15 mL of water into the flask, adding one crystal of manganese dioxide; then he quickly put the stopper into the flask. He measured the volume of gas generated as the water was pushed out of the measuring cylinder. He also noticed that the manganese dioxide was not used up. For test 2, he did exactly the same but left out the manganese dioxide.



a Predict which test produced the most gas the fastest.

b What term do chemists use for the group of chemicals that behave like potassium iodide?

c Explain how this group of substances works.

★ Fluids behaving badly

Newtonian fluids react to temperature but not to shaking (force). Make up a non-Newtonian fluid: a 4% solution of sodium borate (add 4 g borax to 96 g water). Encourage students to arrive at the observation

that if you add a force it behaves like a solid, but when you release the force it behaves like a liquid. This fluid should be prepared just before the challenge – it cannot be made in advance.

★ The challenge

Zygoop is purple and weird but always a surprise. Encourage students to use scientific thinking skills and apply knowledge to determine the properties of this unknown substance, then discuss the results and apply them to an unknown situation. Students are not required to work out what Zygoop is made of (many will guess anyway, but don't admit what it is until the end of class). There are several theories explaining just why cornflour slime and other non-Newtonian fluids (blood, tomato sauce, PVA slime) behave the way they do. When you mix, push, press, or squeeze these fluids, you increase the pressure, which in turn increases the attraction between the starch molecules and the water molecules. This causes the mixtures to act like solids. When the pressure is released the mixtures become runny – the attractive forces reduce and they act like liquids. Adding energy causes Zygoop to become hard on top and gel-like on the bottom. Take energy away and it becomes hard as well. These properties are important in designing a landing craft.

Use newspaper to cover surfaces or set each team up outside in a shady grassed area. To prepare the Zygoop, add drops red and blue food colouring to 1 L of water, producing a strong purple colour. Add 1 kg of cornflour and another 600 mL of water. About 15 minutes before use, 'mix' by hand or use a tablespoon. Lift the mixture, bit by bit, from the bottom of the container and 'flop' it over rather than using 'normal' mixing techniques. The material should flow when you tip the container but feel like a solid when you hit it or rub your finger across the surface. If it is too thick to flow, add a little more water. If it is too soupy, add more cornflour. This is enough Zygoop for six teams to have about 350 mL each. To dispose of Zygoop, scrape as much as possible into a bin, then wash the containers and tools with water. Do not pour it down the drain or it may clog the pipes.

LINKS WITH THE EVERYDAY – spreading butter on bread, silly putty, PVA slime, blood, sand, tomato sauce, stirring canned soup or paint, lava, toothpaste.

★ Curiosity trap

Have students list the properties of a substance they know well, such as ice-cream. Next, take a pot of cheap hair gel and let each student scoop some out with their finger. Ask them to rub it between their fingers – does it become thicker or thinner? It becomes thinner because it is a stir-thinning fluid. Why is this particular physical property necessary for hair gel?

Because it must become thin enough to coat each hair to which it is applied. Lastly, take small bottle of the best quality tomato sauce. Remove the lid and tip it upside down as if attempting to pour it onto a meat pie. Because it's a stir-thickening fluid, it won't come out until you shake it vigorously. Refer to the 'So what's going on?' section for more information.

★ Answers

- 1 Solids: chalk, cotton wool, rubber bands;
Liquids: glass, shaving foam, soft drink, perfume, syrup; Gases: shaving foam, soft drink, air
- 2 **a** solid **b** gas **c** liquid
- 3 **a** add **b** add **c** subtract **d** subtract **e** add
- 4 **a** flows; slide; friction; viscosity **b** High viscosity: honey, engine oil, syrup, cooking oil; Low viscosity: water, milk, fruit juice, soft drink
- 5 **a** colour, solubility, density, melting point **b** Physical properties can be investigated without changing the composition of the substance
- 6 **a** Newtonian **b** The molecules move further apart and move a little faster if energy is added, allowing them to slide over one another more easily
- 7 Stir-thinning: toothpaste, ink in pen, paint;
Stir-thickening: PVA glue, tomato sauce, silly putty
- 8 Magma contains dissolved gases. If the bubbles can rise easily through a low viscosity magma, the lava will flow slowly. If the bubbles get trapped in a

thick magma, they will expand explosively when the pressure is released

- 9 **a** It behaves like a solid and appears to be firm and dry **b** It behaves like a liquid and their feet sink into it **c** A small amount of water separates the individual sand grains and acts as a lubricant. When we just stand on wet sand, the grains under our feet are able to slide past each other out of our way and we sink in. If we jump on wet sand, the water between the grains gets squeezed out and cannot return to fill the cavities fast enough, so the grains get jammed up against each other and friction stops them from sliding out of our way – the sand mixture behaves like a solid **d** a non-Newtonian fluid
- 10 **a** changes in the oil due to temperature **b** Yes, because if an oil reacted to the force exerted on it by the moving parts then it might either become so thin that frictional wear would occur or so thick that the parts would not slide easily, slowing down the engine

☆ Give us a clue – fluids behaving badly

What you need

Experiment 1

5 test tubes, filled with the following fluids: milk, cooking oil, glycerine, honey, water; 15 paperclips; stopwatch; access to a hot water bath

Experiment 2

Ziploc bag; white PVA glue; food colouring; borax solution; 20 mL measuring cylinder; scissors

What to do

Experiment 1: Newtonian fluids

- 1 **Drop a paperclip** into each fluid (one at a time) and use the stopwatch to carefully time how long it takes to reach the bottom of the tube.
- 2 **Put your thumb** over the end of each tube and shake vigorously.
- 3 **Repeat step 1**, again noting the times.
- 4 **Place the test tubes** into the hot water bath for 5 minutes, then take them out and repeat step 1 again.

Experiment 2: Non-Newtonian fluid

- 1 **Take** a ziploc bag and add 20 mL of PVA glue, 20 mL of water and 4 drops of food colouring. Seal the bag firmly and mix by squishing and kneading the mixture in the bag.
- 2 Now **add** 15 mL of borax solution, close the bag firmly and knead the bag between your fingers for 10–15 minutes.
- 3 **Take** the fluid out of the bag and knead it some more in your fingers, forming it into a ball. Observe what happens when you: set it on a piece of paper, hit it hard, stretch it, walk over it slowly with your fingers, run over it with your fingers, bounce it, and cut it with scissors.



Now answer these questions:

- Which fluid in Experiment 1 is the most viscous and which is the least viscous?
- Did shaking or temperature cause any change in these fluids?
- When you held the fluid in Experiment 2 in your hands, how did it behave?
- Is it a solid or a liquid?
- What are the properties of this non-Newtonian fluid?

What happens next?

After you have completed these experiments and answered the questions above, you will be asked to solve a problem. You will need to use the information learned from this experiment and apply it to the Challenge that follows. So watch, listen, look for clues and stay alert!

Clues!

All fluids have an internal friction, known as **viscosity**, which makes them resist flowing.

The viscosity of **Newtonian fluids** is changed by varying the temperature.

The viscosity of **non-Newtonian fluids** can be changed by temperature or by adding a force (applying pressure).

★ The challenge

Determine the properties of Zygoop, then design a landing craft that could land on Zygoop and take off again without sticking, falling over or sinking.

Materials

350 mL of Zygoop in a 2 L ice-cream container; access to a hotplate and a freezer; two small beakers; a foam disposable cup containing a metal teaspoon and the following materials: a paddle-pop stick, paper (5 cm × 5 cm), a cork, bubble wrap (5 cm × 5 cm), aluminium foil, plastic wrap

Instructions

We do not know what Zygoop is made of, but a team of scientists are working this out. All will be revealed when your research is complete. We do know that Zygoop is safe to handle.

- 1

Using the materials provided, work out how Zygoop behaves.
- 2

Through discussion with your team, decide which of the properties you have identified will need be taken into account when designing the landing craft. List these properties in Box 1 below.
- 3

In Box 2, draw a sketch of the landing craft and label the materials used for each component.
- 4

Show the judge your list of the relevant properties of Zygoop and explain how you have incorporated these properties into your landing craft design.

SAFETY!

⚠ We do not know what this mysterious substance is made of, so do not try to taste it.

BOX 1: PROPERTIES OF ZYGOOP	BOX 2: LANDING CRAFT DESIGN

Clues!

Some substances can behave as both a solid and a liquid.

A fluid that gets thicker if a force is added to it (stirring, shaking, slapping, squeezing)

is known as a **stir-thickening** fluid – its molecules don't flow easily under pressure.

Stir-thinning fluids get thinner and flow easier when a force is added to them.

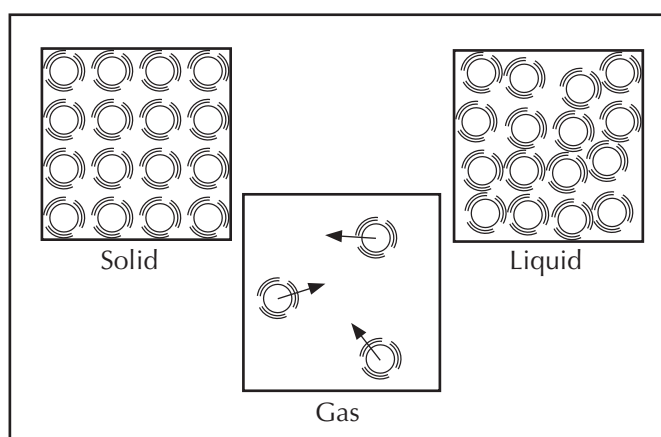
★ So what's going on?

Invisibly small pieces

We live in a vast and lonely universe, filled with matter that we cannot yet identify and whose properties and physical laws we do not fully understand. However, scientists believe that the same amount of matter has been around since the beginning of time. The matter on Earth exists in three states: solids, liquids and gases. The difference between these groups is how closely packed together their molecules are. This in turn depends on how much they are moving. How much they are moving depends on how much energy they have. To change the state of matter, energy can be either added or removed – but the substance will still be made of the same molecules. All three states of matter have mass, volume and density.

The way things are

In a solid, the atoms are packed closely together, vibrate on the spot and maintain their shape. In a liquid the atoms vibrate enough and have enough space between them to slide past one another. A liquid maintains its volume (the amount of space it takes up), can be poured, and takes the shape of its container. In a gas, the atoms fly around and collide with other atoms. A gas has no fixed volume and has the most energy; it therefore takes up the volume of whatever container it is in.



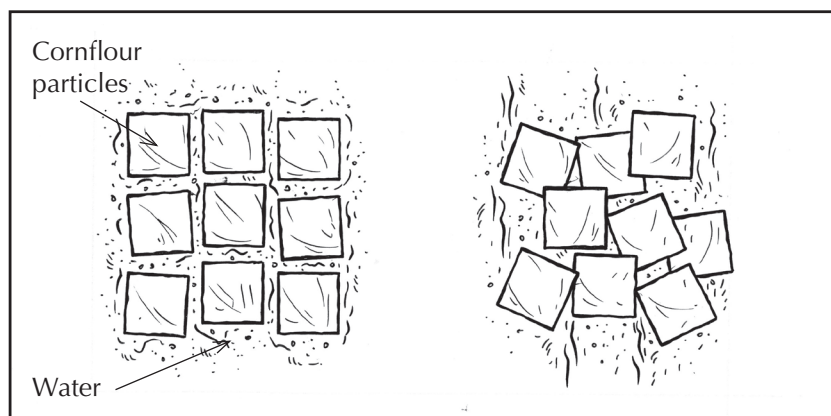
Obeying the rules

Solids, liquids and gases have different properties. The properties of a substance are the qualities that make it different from other substances and determine the ways it can be handled or used. We identify and classify substances and objects based on the specific set of properties they possess.

The **physical properties** of a substance can be investigated without changing its composition. Some examples of physical properties are colour, shape, mass, density, hardness, boiling and melting points, odour, viscosity. The **chemical properties** of a substance are the ways in which its atoms react with other atoms.

Fluids that do not behave

Fluids are anything that flow, including gases, liquids, mud flows and glaciers. To flow, the molecules must slide over one another while the friction between them resists this movement. How thick a fluid is, or how much it resists flowing, is called its viscosity. Fluids with a high viscosity are usually thick and they pour slowly (like honey). Fluids with a low viscosity are thin and flow faster (like water). All fluids



have this physical property known as viscosity. Sir Isaac Newton's Law of Viscosity says that a fluid's viscosity can be changed only by altering the fluid's temperature. For example, if you warm up honey or oil (adding heat energy) it flows more easily, and we say that it becomes less viscous. Newtonian fluids obey the rules and their viscosity remains constant at a constant temperature. However, the viscosity

of **non-Newtonian fluids** can also be changed by applying a force. Added force (stirring, squeezing, compressing) either increases or decreases the viscosity of these fluids, so they do not obey Newton's rules when under pressure.

One theory that tries to explain a stir-thickening fluid like cornflour and water is that pressure causes the cornflour molecules to jam up, so they cannot slide smoothly and the mixture feels hard. When you relax the force, they can slide again.

The last word

Stir-thickening fluids are being considered for body armour as they absorb the energy of a speeding projectile but remain flexible when hit by slow-moving objects.

★ **Figure it out!**

- 1 Tick which group the substance belongs to – some belong to more than one group and some are just plain tricky!

SUBSTANCE	SOLID	LIQUID	GAS
Glass			
Shaving foam			
Soft drink			
Perfume			
Syrup			
Chalk			
Cotton wool			
Air			
Rubber bands			

- 2 Label the columns with the correct state of matter.

Particles are:

	A. _____	B. _____	C. _____
Packed	closely	spaced out	closely
Arranged	regularly	at random	irregularly
Moving	on the spot	very fast	past each other

- 3 The table below lists five different changes of state. For each change, decide whether energy must be added or subtracted. Tick the correct columns.

CHANGE OF STATE	ADD ENERGY	SUBTRACT ENERGY
a solid to liquid		
b liquid to gas		
c gas to liquid		
d liquid to solid		
e solid to gas		

- 4 Read the following information about the behaviour of fluids and fill in the missing words.
- a** Fluids are anything that _____. To flow the molecules must _____ over one another but the _____ between them causes a resistance to how well they flow. How thick a fluid is, or how much it resists flowing is called its _____.
- b** Group the following fluids into two groups – ‘high viscosity’ and ‘low viscosity’ – honey, water, engine oil, milk, fruit juice, soft drink, syrup and cooking oil.
- 5 Every substance has both chemical and physical properties.
- a** Which of the following are physical properties: colour, ability to burn, solubility, density, melting point, production of a gas?
- b** Explain your answer.
- 6 If energy is taken away from honey it becomes thicker. If we add energy it becomes thinner.
- a** Based on these properties, what group of fluids does honey belong to?
- b** At a molecular level, how does temperature affect the viscosity of fluids?
- 7 The following table lists some fluids that are affected by pressure. Tick the substances that can be considered stir-thinning and those that are stir-thickening.

FLUID	STIR-THINNING	STIR-THICKENING
Toothpaste		
Ink in a ballpoint pen		
PVA glue		
Tomato sauce		
Silly putty		
Paint		

- 8** A group of students are investigating the viscosity of several fluids. They decide to assess viscosity by recording how long it takes for air bubbles to move to the surface of the fluids. They find that in thin liquids like water, bubbles rise smoothly to the surface. In thick liquids like syrup and glycerine the bubbles have difficulty, and when they finally reach the surface they burst through and explode with a loud pop.

Read the following information about volcanoes and relate the movement of bubbles through a liquid to these two types of volcanic eruptions.

Some erupting volcanoes are quiet and the lava flows smoothly from the vent and down the mountainside. Violent eruptions occur when the lava is thicker and contains a lot of gas bubbles, which explode from the vent when the pressure on them is released.

- 9** When visiting the beach you often see children jumping up and down on the wet sand near the water's edge. You can also see adults twisting their feet as they search for shellfish to use for fishing bait.

a As they move their feet, what happens to the sand?

b When they stop moving their feet, what happens to the sand?

c What is one theory to explain this strange phenomenon demonstrated by fluids such as sand and water?

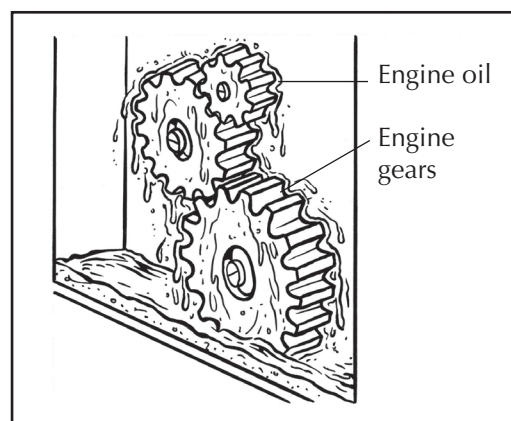
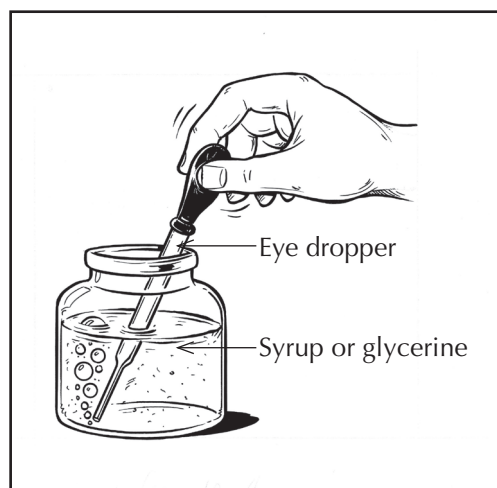
d Which type of fluid is a mixture of sand and water?

10 Extend your understanding

We use lubricants such as engine and transmission oils in vehicles to reduce the friction between moving parts. There are many different grades of motor oil and these are classified by industry codes called 'viscosity indexes'. These are written on the containers in a form such as '20/50W'. The first number is the nominal viscosity at low temperatures and the second at higher temperatures. As long as the viscosity of the lubricant remains at the specified level, the surfaces can slide relative to one another without causing mechanical wear. In the modern car industry, the viscosity of lubricants is a very exact science.

a Which physical property of fluids must be considered when producing a 'multi grade' lubricant for use in all climates?

b Would stir-thinning or stir-thickening properties also need to be considered when designing motor oils? Explain your answer.



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